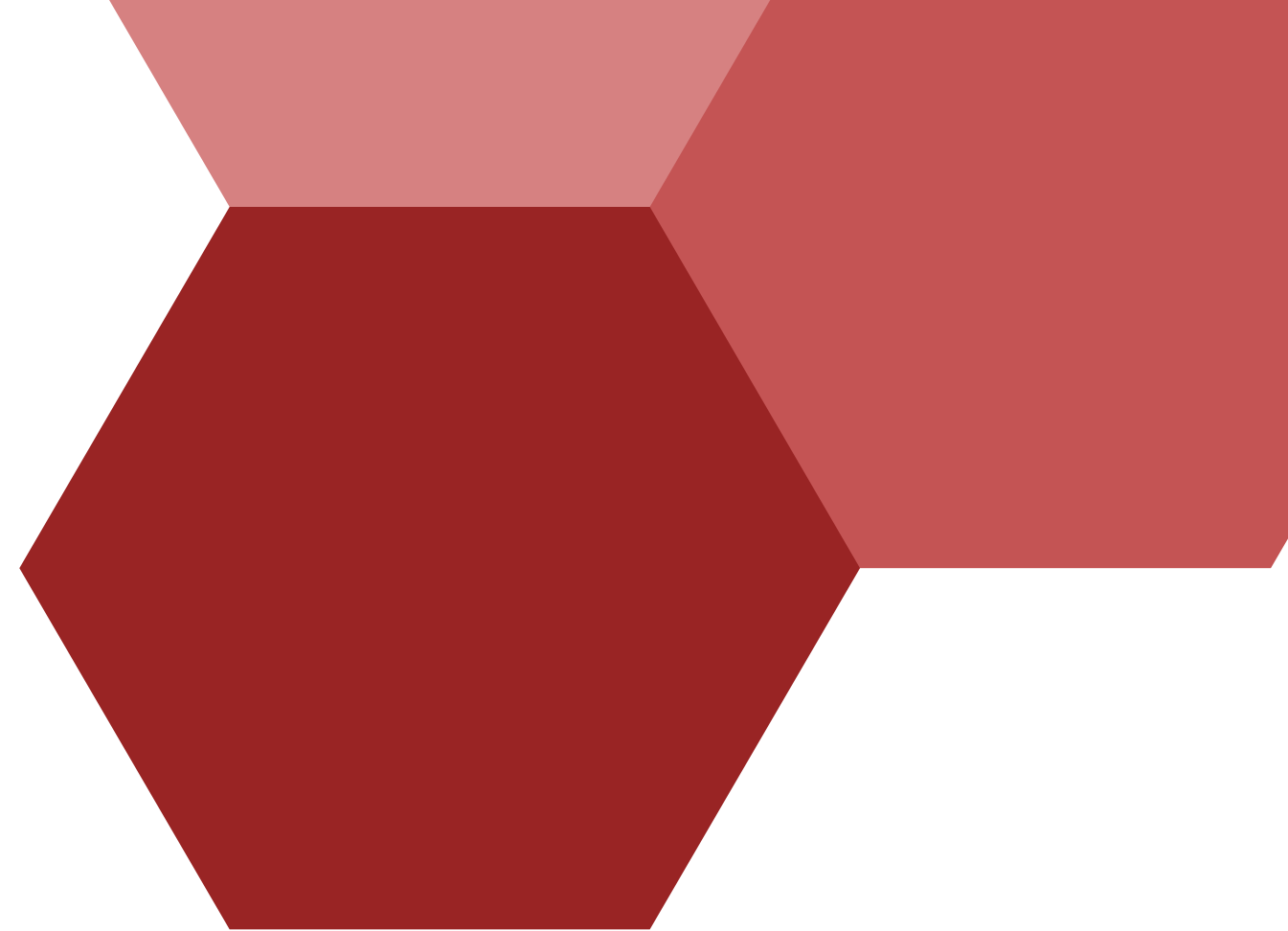


CHAPTER 3

HARDWARE



CHAPTER OUTLINE

3.1
Overview of
computer system
hardware
functionality

3.2
Embedded
system

3.3
Memory
components

3.4
Secondary
storage
devices

3.5
Output
devices

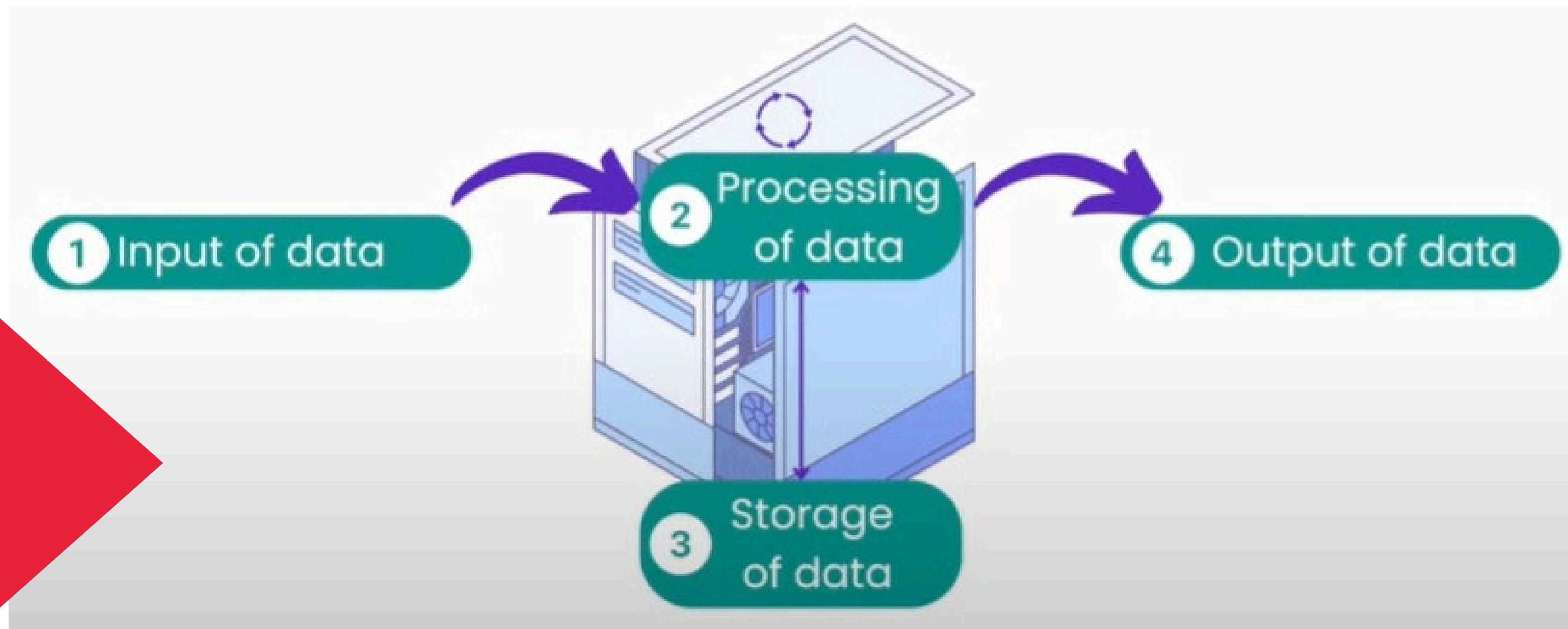
3.6
Input
devices

3.7
Input and output of
sound

3.1 Overview of computer system hardware functionality

COMPUTER SYSTEM

A computer system has to support the following operational capability:



3.1 Overview of computer system hardware functionality

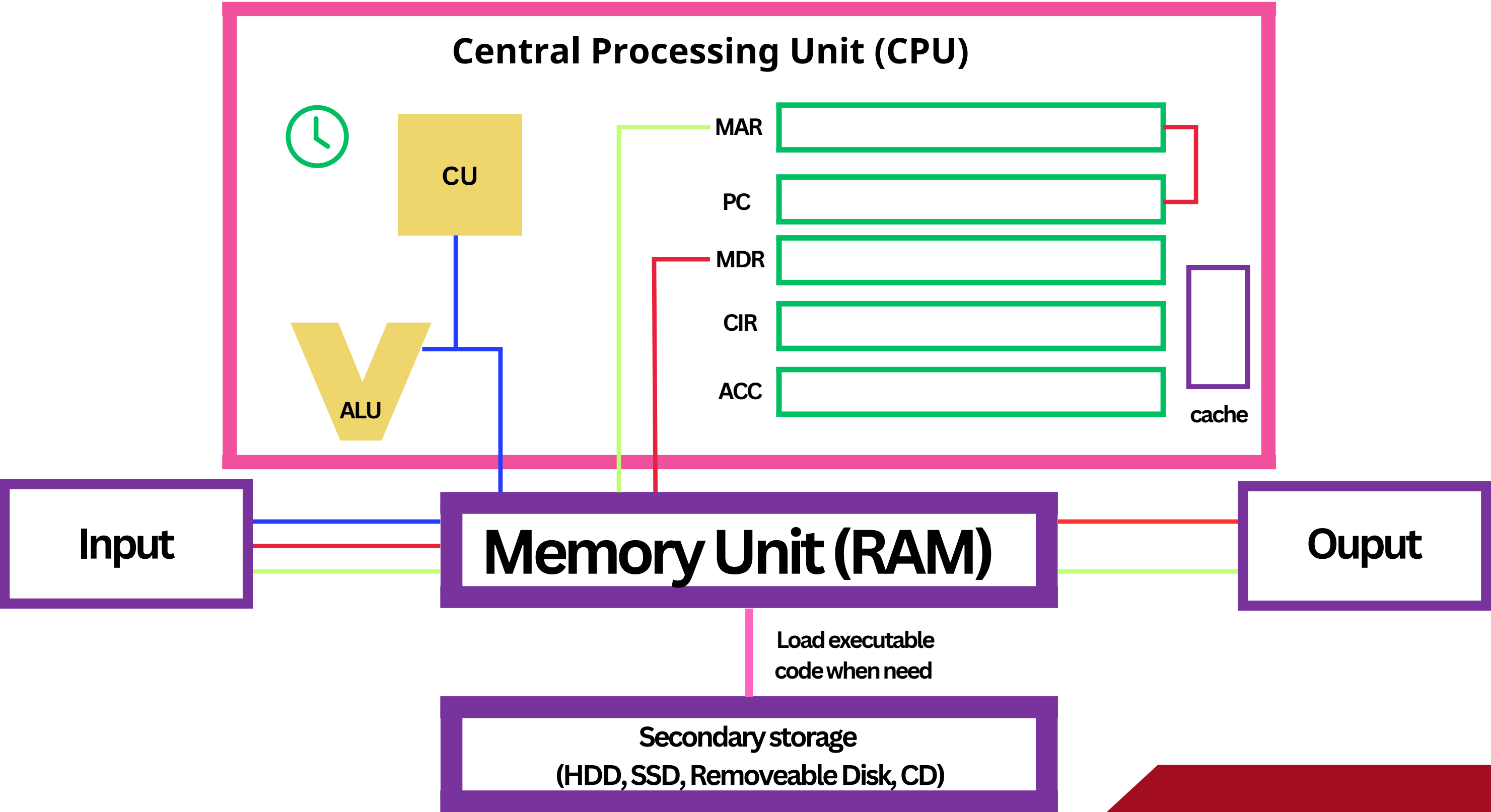
VON NEUMANN ARCHITECTURE

Control Bus

Data Bus

Address Bus

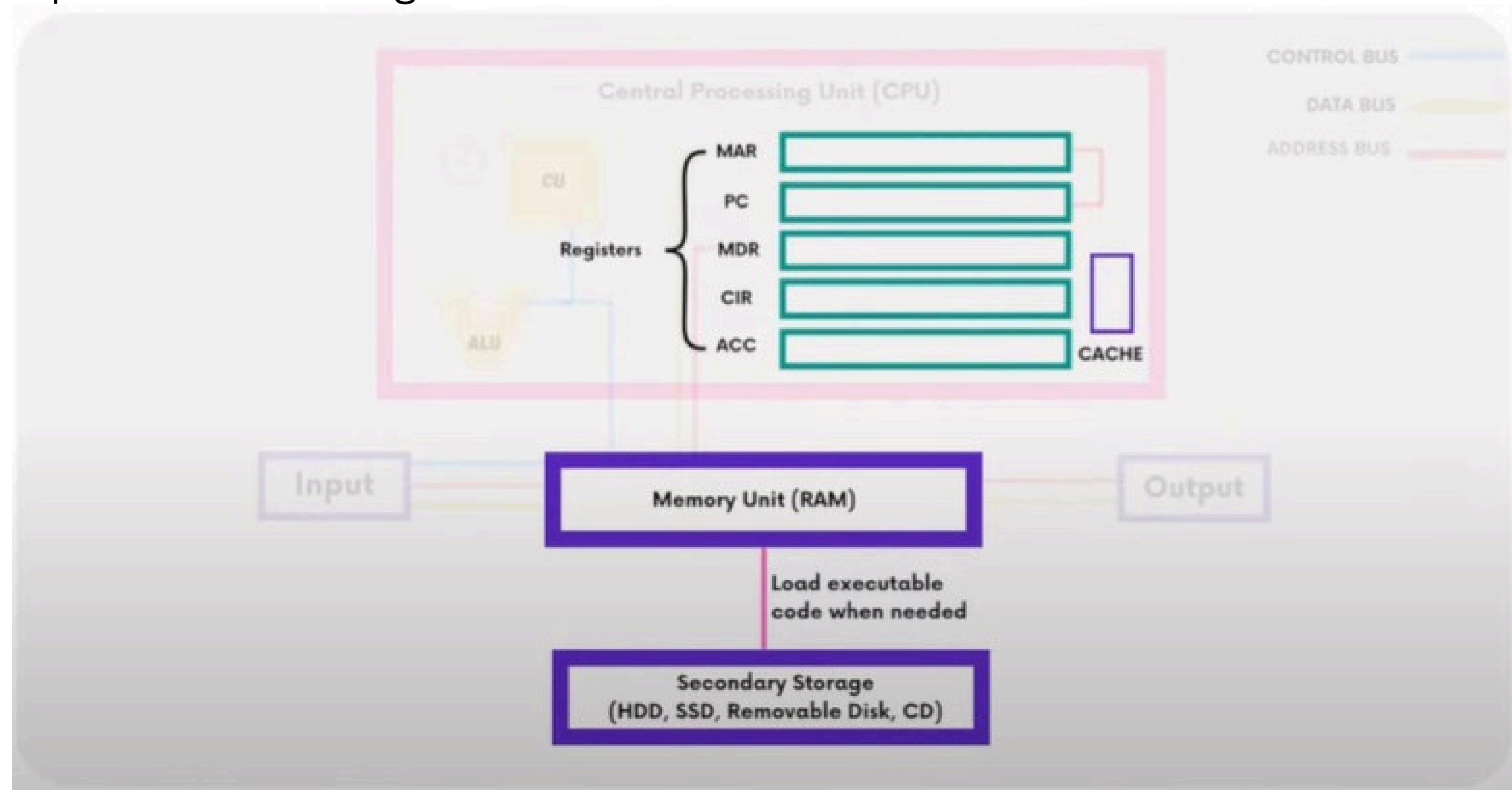
The von Neumann architecture is a computer design that incorporates a single **memory** to store both data and instructions, with a **central processing unit (CPU)** that can sequentially fetch, decode, and execute these instructions.



3.1 Overview of computer system hardware functionality

Storage of Data.

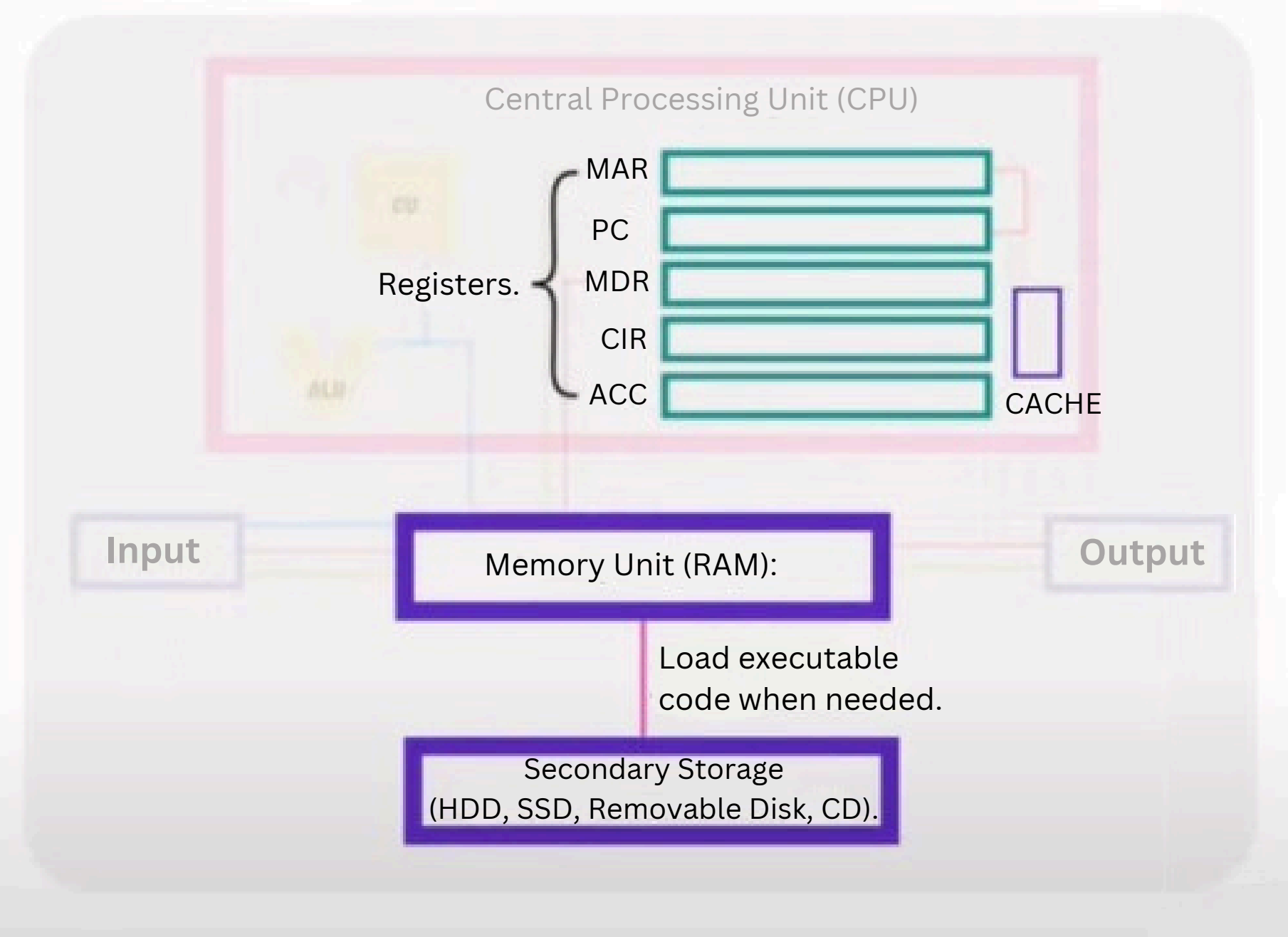
The following are the components for storage of data.



3.1 Overview of computer system hardware functionality

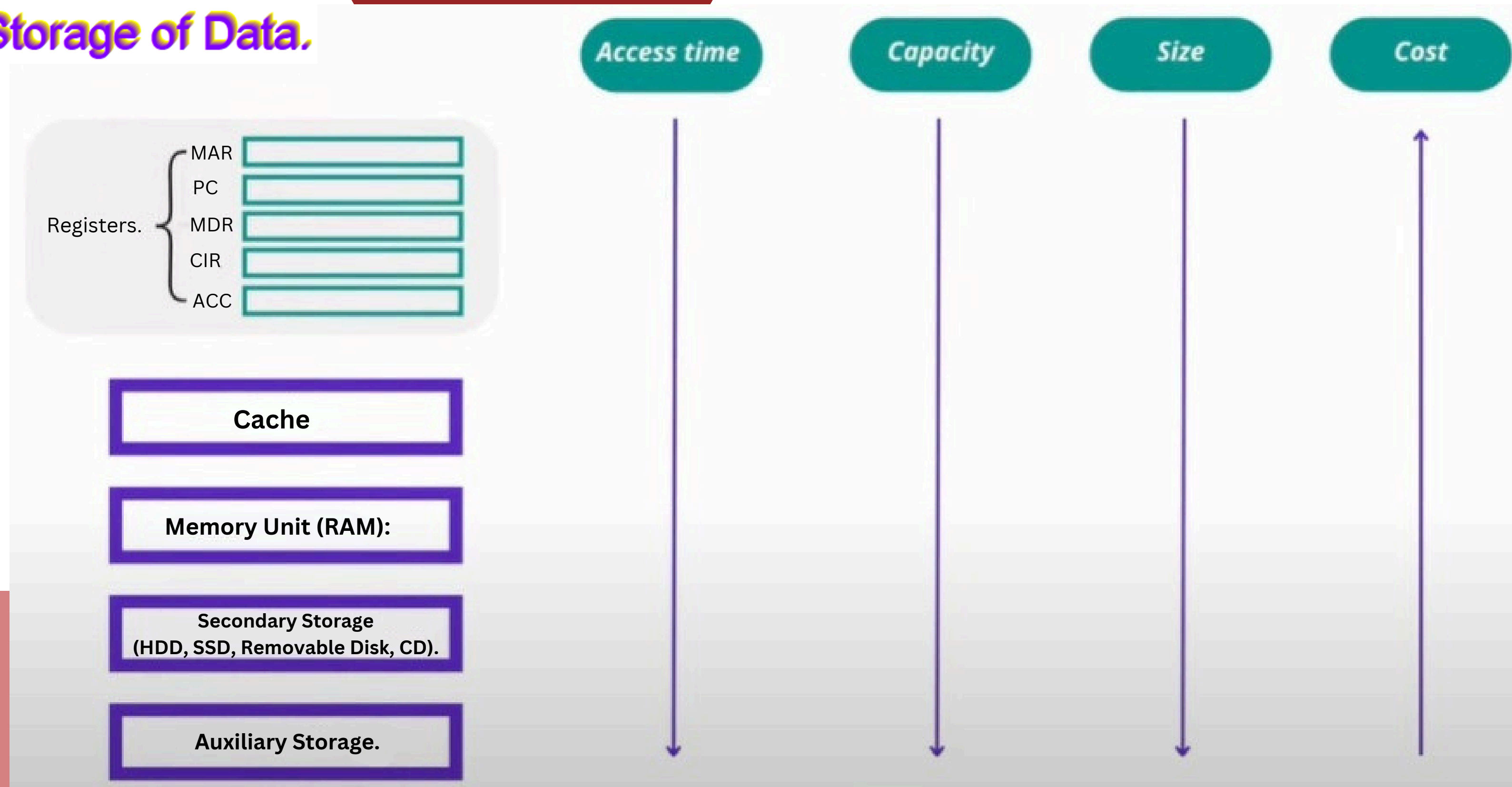
Storage of Data.

The memory hierarchy serves as a valuable concept when deciding on the elements to employ in a computer system for storing data.



3.1 Overview of computer system hardware functionality

Storage of Data.



3.1 Overview of computer system hardware functionality

Storage of Data.



Corsair Vengeance RGB Pro
64GB (4x16GB) DDR4 3000
(PC4-24000) C15 Desktop

5.0 ★★★★★ 2

Four

RM 2,454.94 + tax (US\$517.81 + tax)



Vaseky v800 SOLID STATE DRIVE
SSD 128GB 256GB 500GB 1TB
2.5" SATA

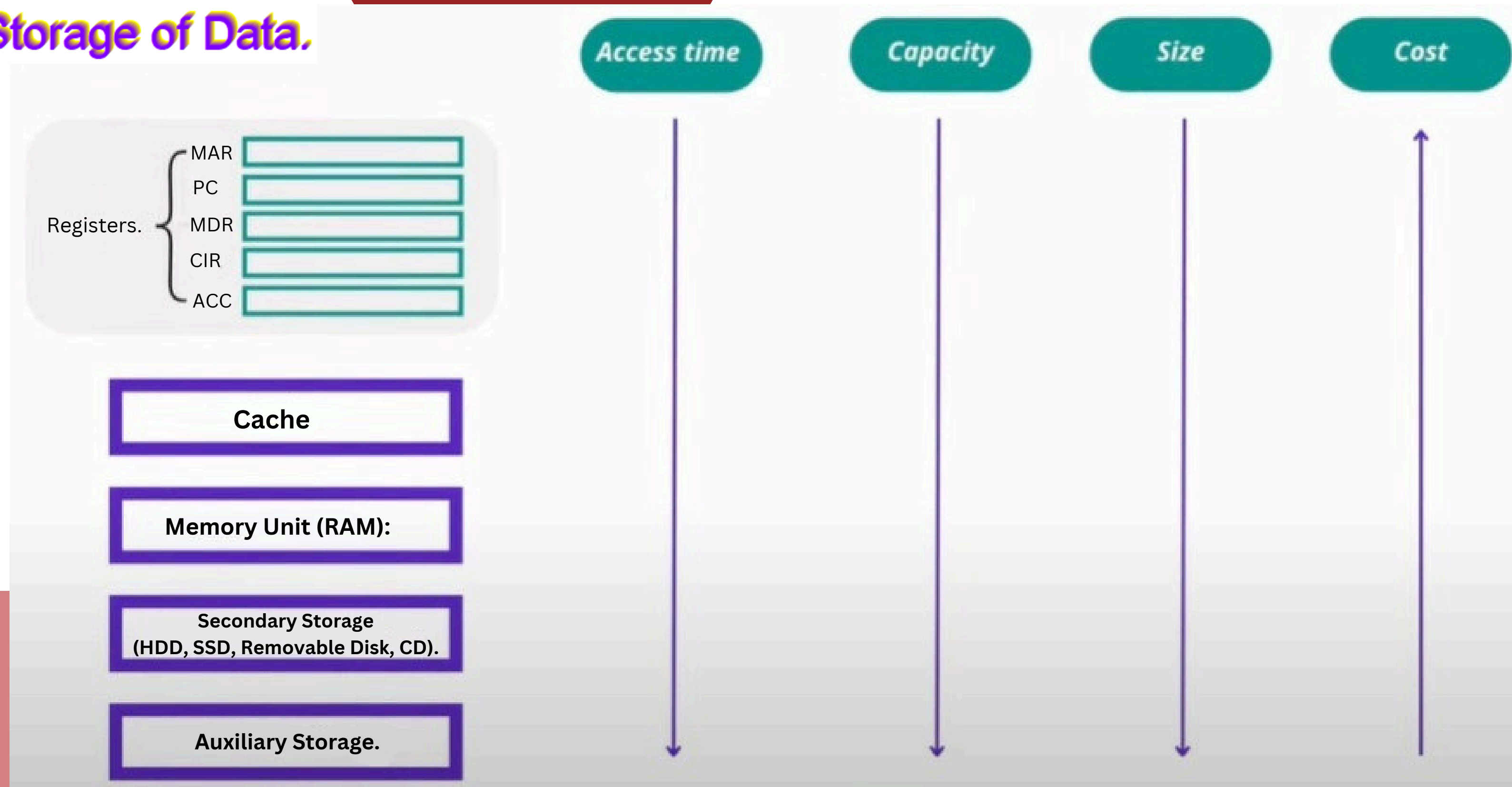
Internal · SATA

RM 90.00

[Shopee](#)

3.1 Overview of computer system hardware functionality

Storage of Data.



3.1 Overview of computer system hardware functionality

Storage of Data.



Corsair Vengeance RGB Pro
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SSD 128GB 256GB 500GB 1TB
2.5" SATA

Internal · SATA

RM 90.00

[Shopee](#)

3.1 Overview of computer system hardware functionality

Storage of Data.

Function vs Hardware.

<i>A storage device that is an integral part of the system</i>	<i>Hard Disk or Solid State Drive</i>
<i>An individual storage object that can be placed into a computer system's drive</i>	<i>Floppy disk, optical disc, magnetic tape cartridge</i>
<i>Peripheral device that can be connected to the system when needed</i>	<i>Hard drive, memory stick/card</i>
<i>Portable memory device that can be carried around</i>	<i>Flash memory stick, floppy disk, optical disc</i>
<i>Remote from the system, accessible via a network connection</i>	<i>Cloud storage, magnetic tape</i>

3.1 Overview of computer system hardware functionality

Storage of Data.



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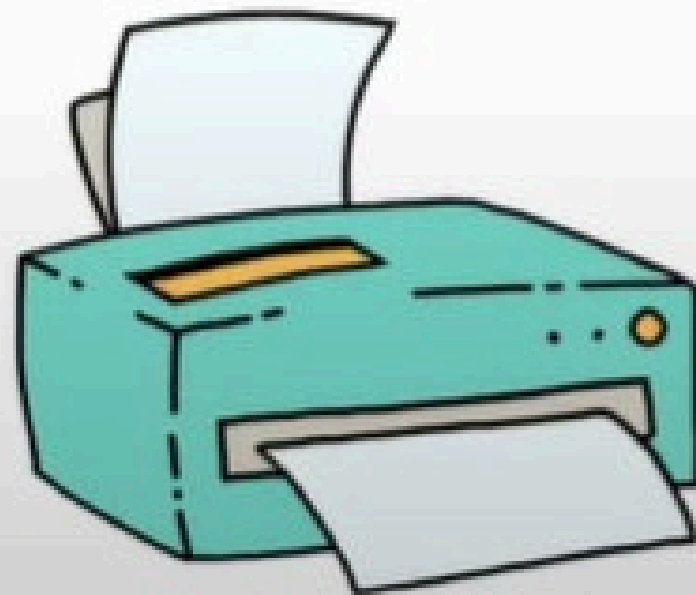
3.1 Overview of computer system hardware functionality

Output device.

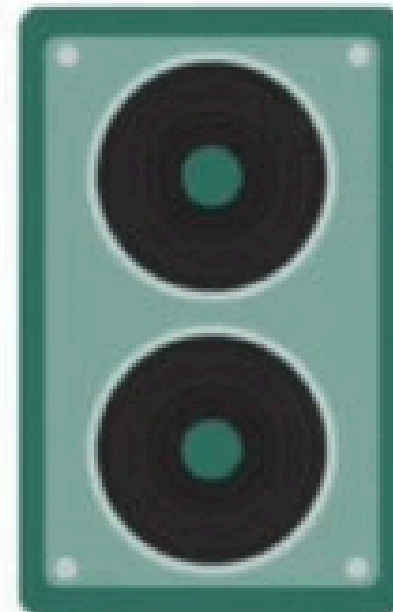
When it comes to data output from a computer system, you have the following choices:



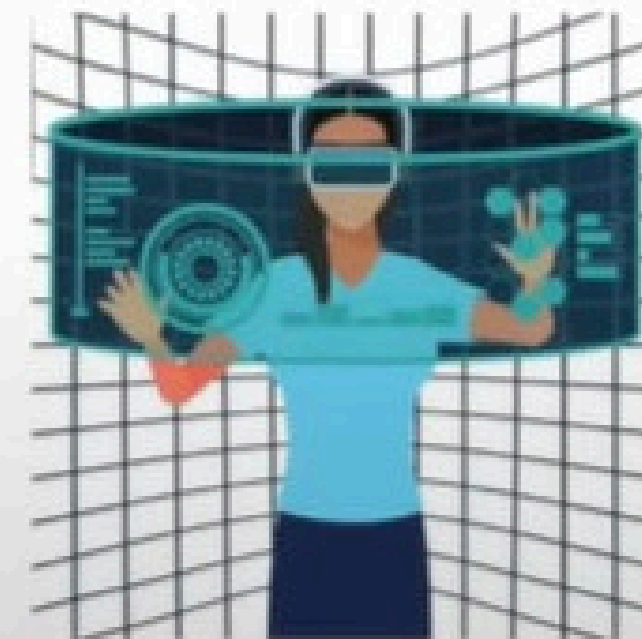
Screen



Printer



Speaker



*Virtual Headset
Display*



*Writing to data
storage device*

etc.

3.1 Overview of computer system hardware functionality

Input device.

When it comes inputting data into a computer system, you have the following choices:

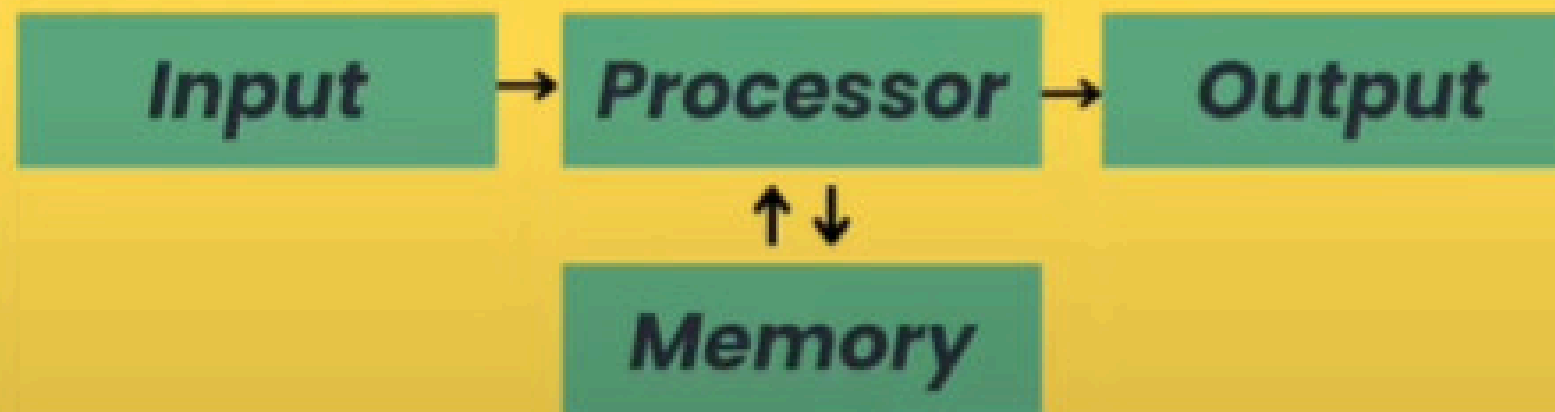


3.2 Embedded system

Definition

An embedded system is a ***specialised computer system*** designed to perform dedicated functions within a larger electronic or mechanical system. Their characteristics:

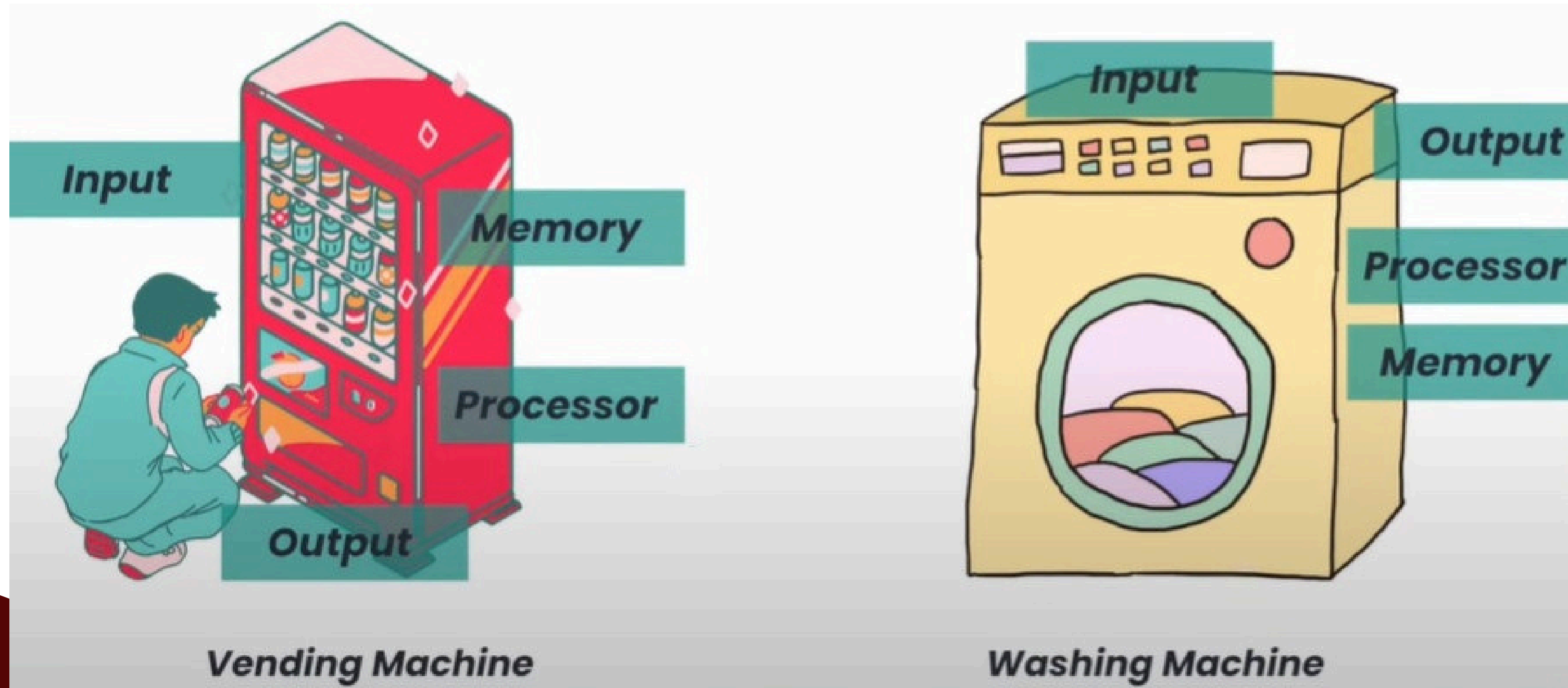
Structure: It must contain a processor, memory and an I/O capability.



Purpose: They are special-purpose, possibly performing only a single function.

**If these systems are built on a single integrated circuit, they are referred to as microprocessors.*

Examples of Embedded Systems



3.2 Embedded system

Advantages

These systems are typically designed for a specific purpose and may specialize in executing just one function.

Due to economies of scale, these chips can be relatively cheap.

Disadvantages

Must

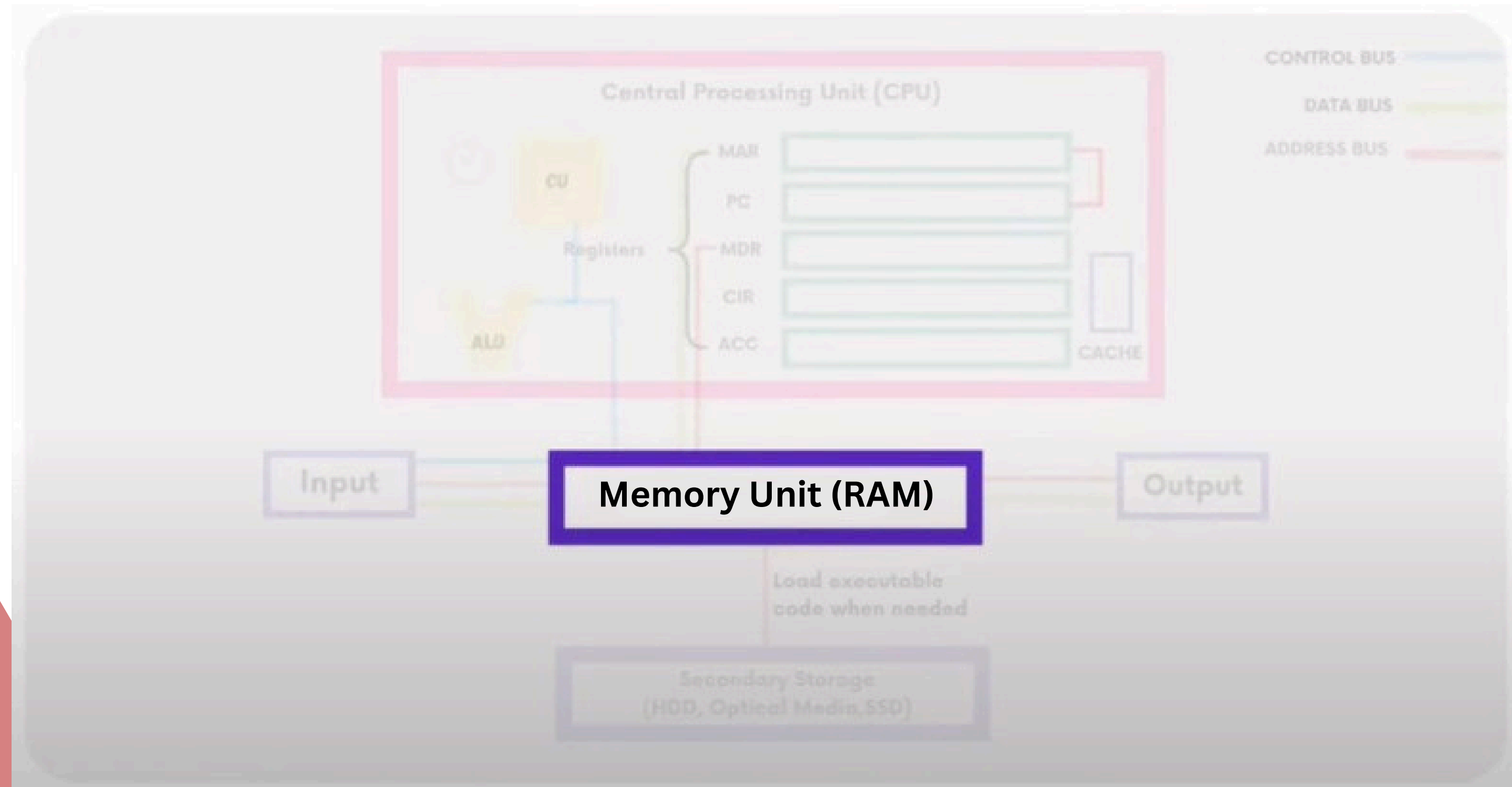
Program has to be short as the memory space available is limited.

Slow

Embedded systems are integral components of the Internet of Things (IoT), and as more of these embedded systems become connected to networks, their security measures may be less robust, making them susceptible to unauthorized actions.

3.3 Memory components

Big Picture

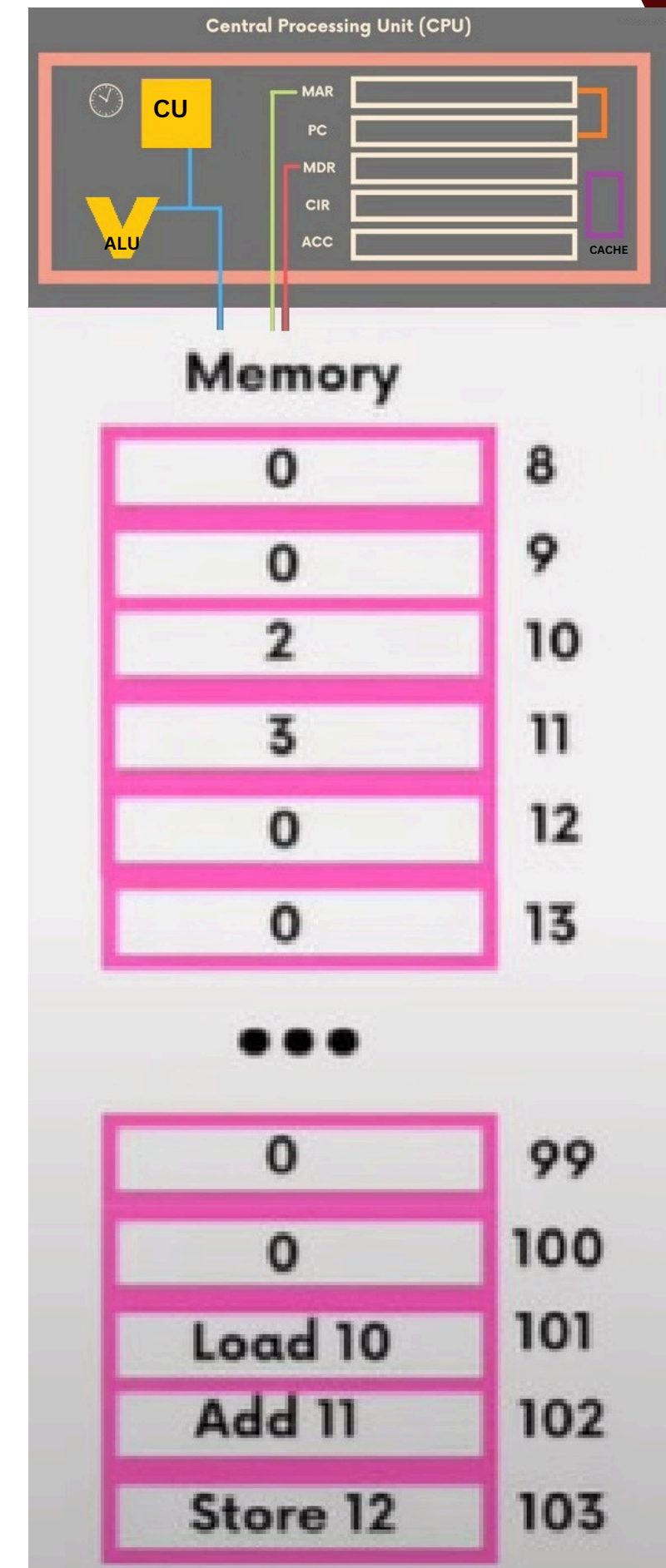


3.3 Memory components

Random Access Memory (RAM)

RAM (Random Access Memory) is a type of computer memory that provides **high-speed data access** to temporarily store and retrieve data that the CPU (Central Processing Unit) can quickly work with during computer operations.

- Also known as direct-access memory
- It loses its stored data when the computer's power is turned off or interrupted.



3.3 Memory components

Types of RAM

- **DRAM (Dynamic RAM)**

is a type of volatile computer memory that stores data in capacitors within cells and requires periodic refreshing to maintain data integrity,

- Cheaper
- Requires fewer electronic components per bit stored

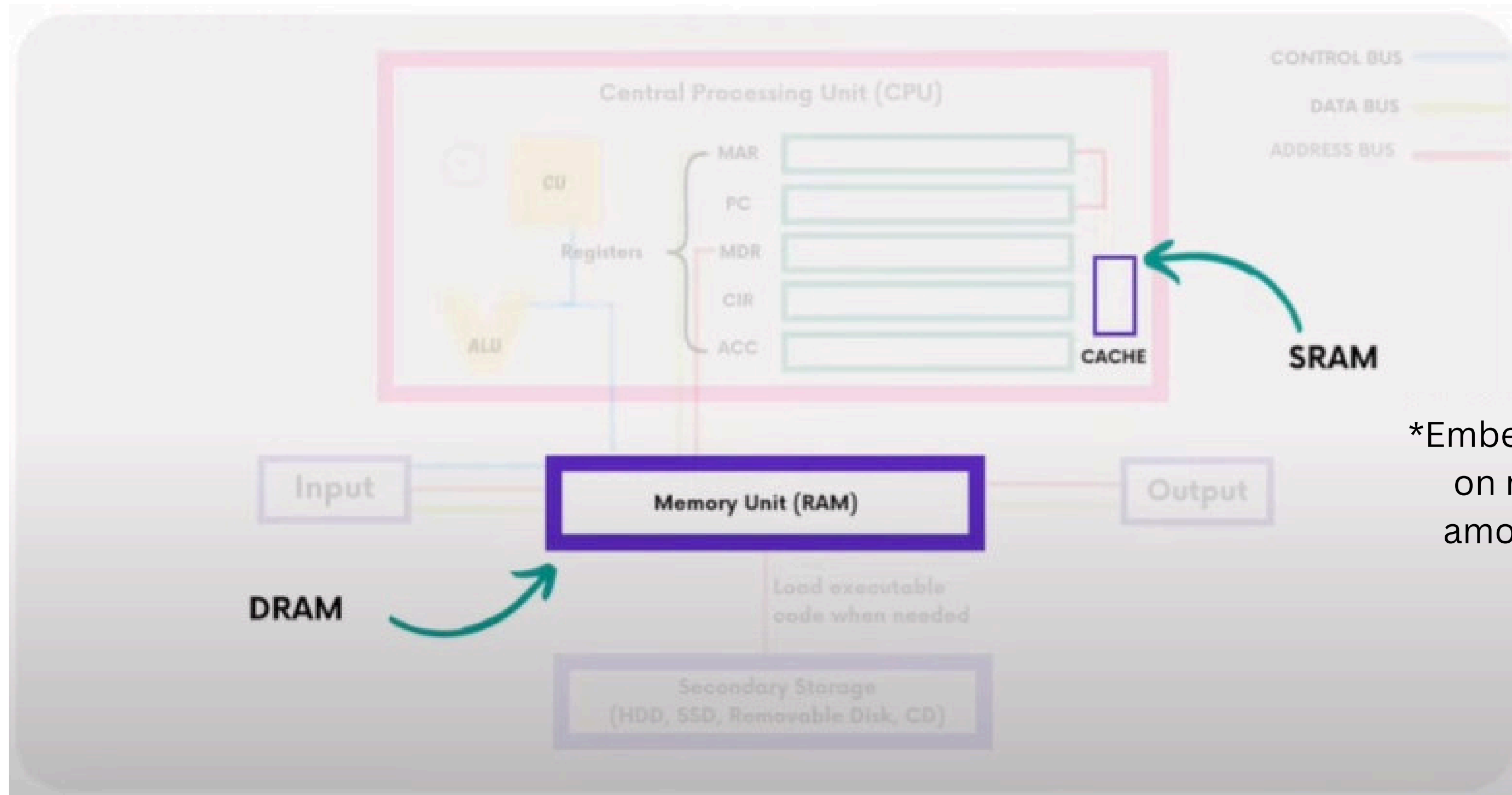
- **SRAM (Static RAM)**

is a type of volatile computer memory that stores data using flip-flops, allowing for faster access and lower power consumption compared to DRAM

- Shorter access time

3.3 Memory components

Types of RAM



*Embedded system that on requires a small amount of RAM also uses SRAM.

3.3 Memory components

Read Only Memory (ROM)

- Shares the random-access or direct access properties of RAM.
- Cannot be written to when in use within the computer system.
- Used to store bootstrap program (startup file). The program run immediately when a system is switched on.



So, while ROM is an important component of a computer system, it is not part of the core von Neumann architecture.

3.3 Memory components

Types of ROM

Simple ROM

- Installed as part of the manufacturing process.
- Chips need to be replaced if different contents are needed.

Programmable ROM (PROM)

- Allow program to be written to by a system builder once.
- After installation, data becomes permanent and cannot be changed or erased.

Erasable PROM

- Data/program in place can be deleted using ultraviolet light.
- A new program can be loaded, but the chip must be taken out of the circuit.

Electrically EPROM

- Electricity is used to remove existing data.
- The chip can stay in the circuit as its contents are modified.

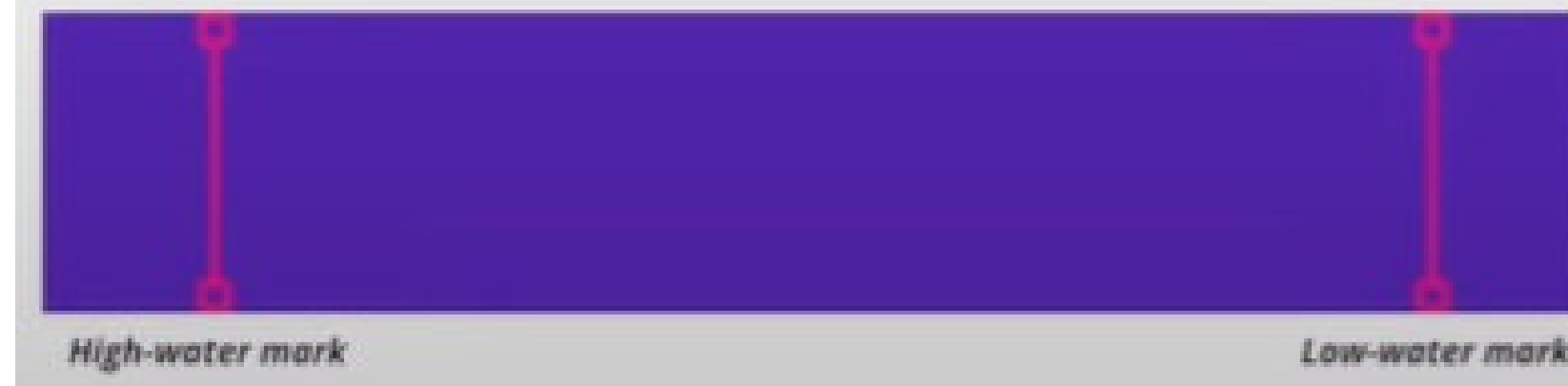
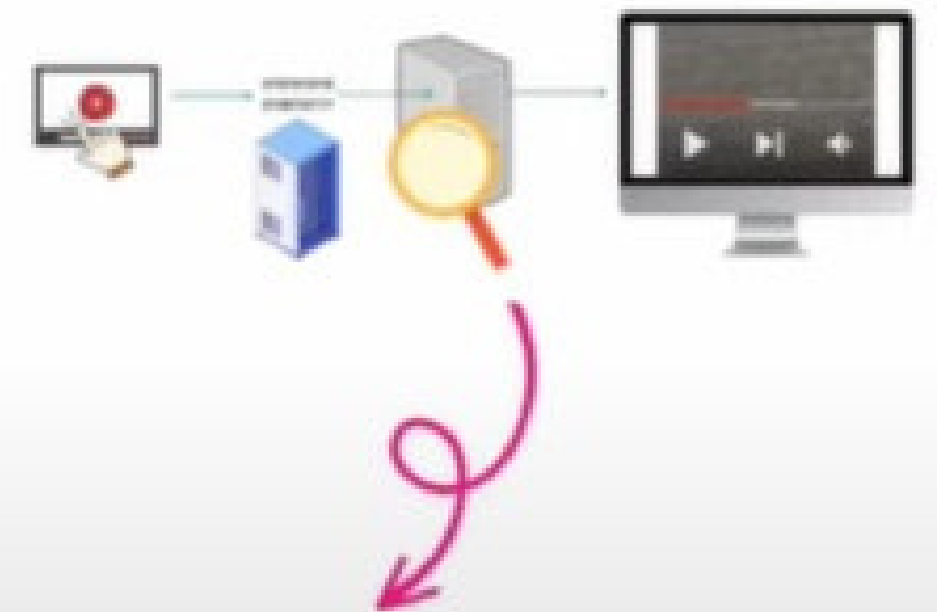
3.3 Memory components

Buffer

When data is sent from one computer to another, problem arises if data is sent more **quickly** than it can be received.

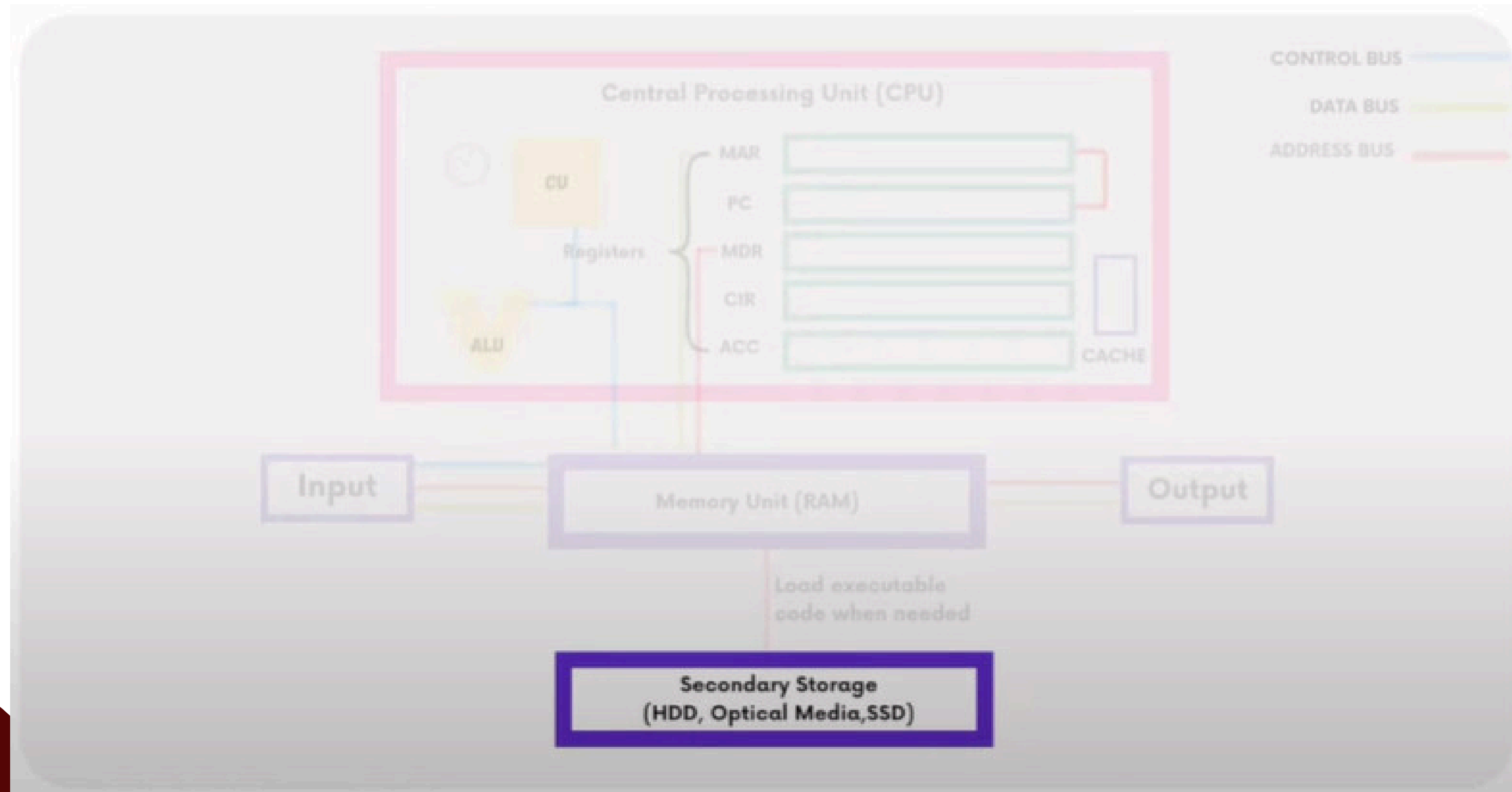
The solution is to use a buffer. A buffer acts as a temporary storage and operates as a queue for data moving between system components.

Typically, buffer is created in the computer memory.



3.4 Secondary storage device

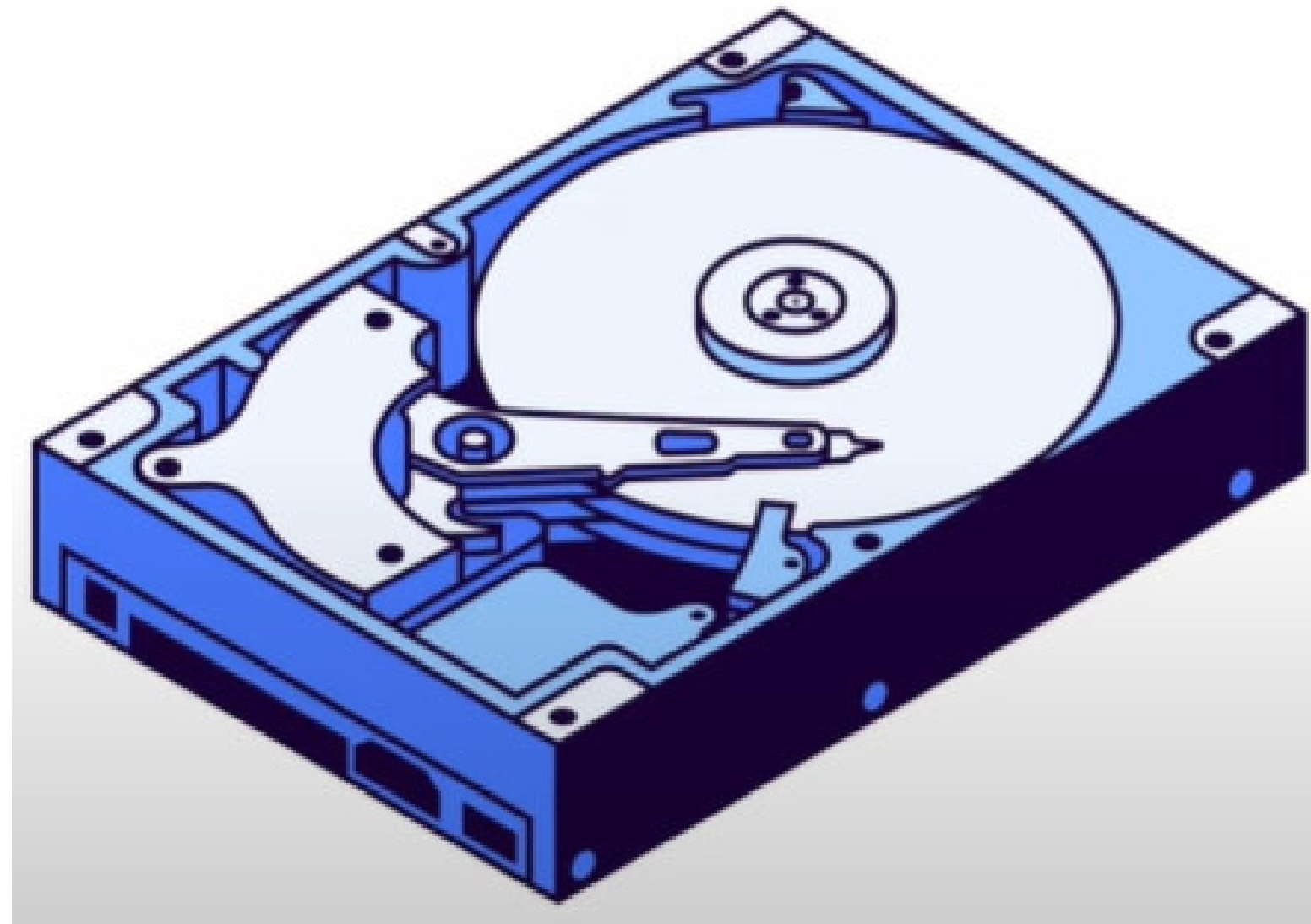
Big Picture



3.4 Secondary storage device

Magnetic Media

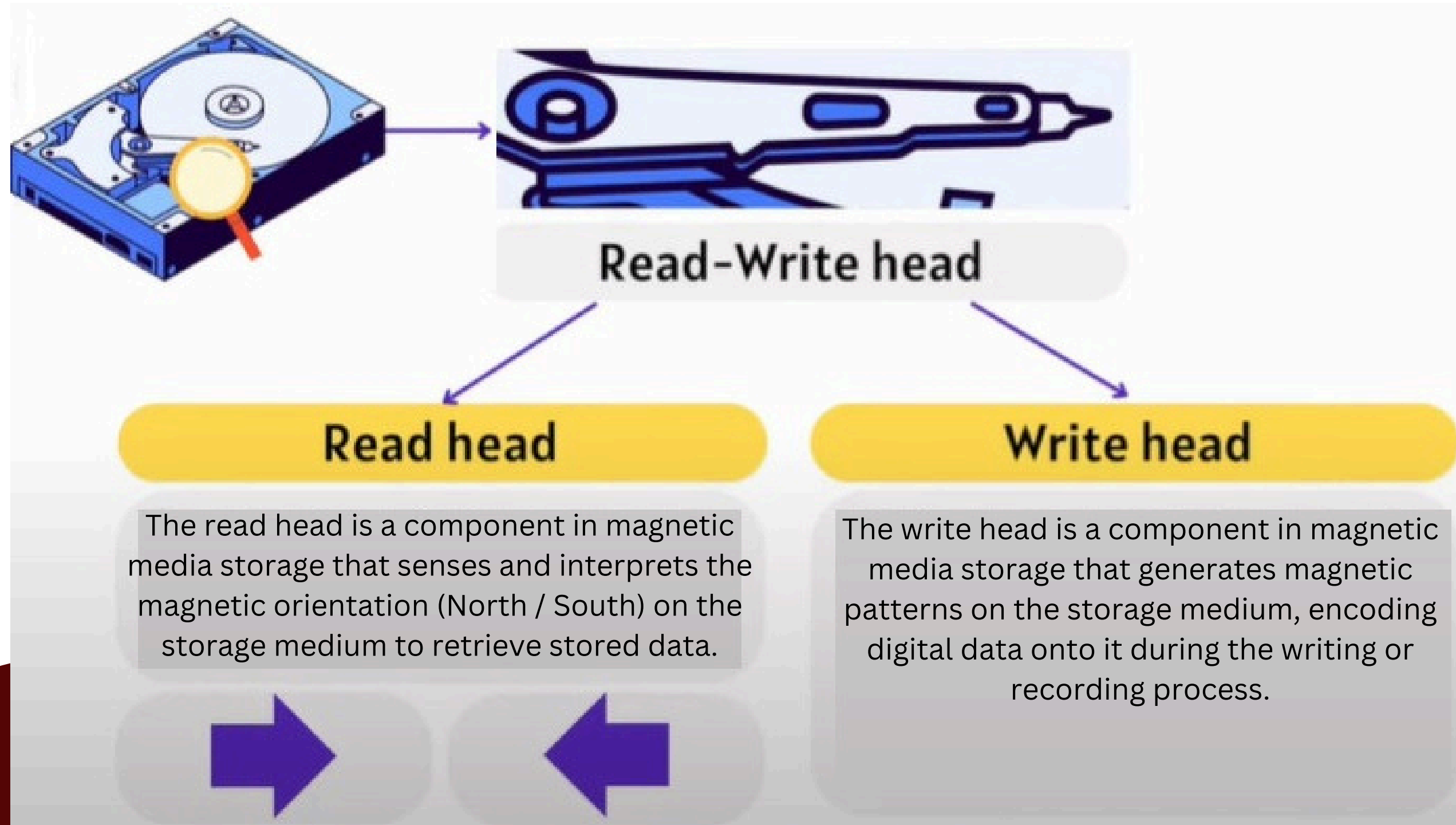
Magnetic media refers to storage devices that use magnetic patterns to encode and store digital data, such as hard disk drives and magnetic tapes.



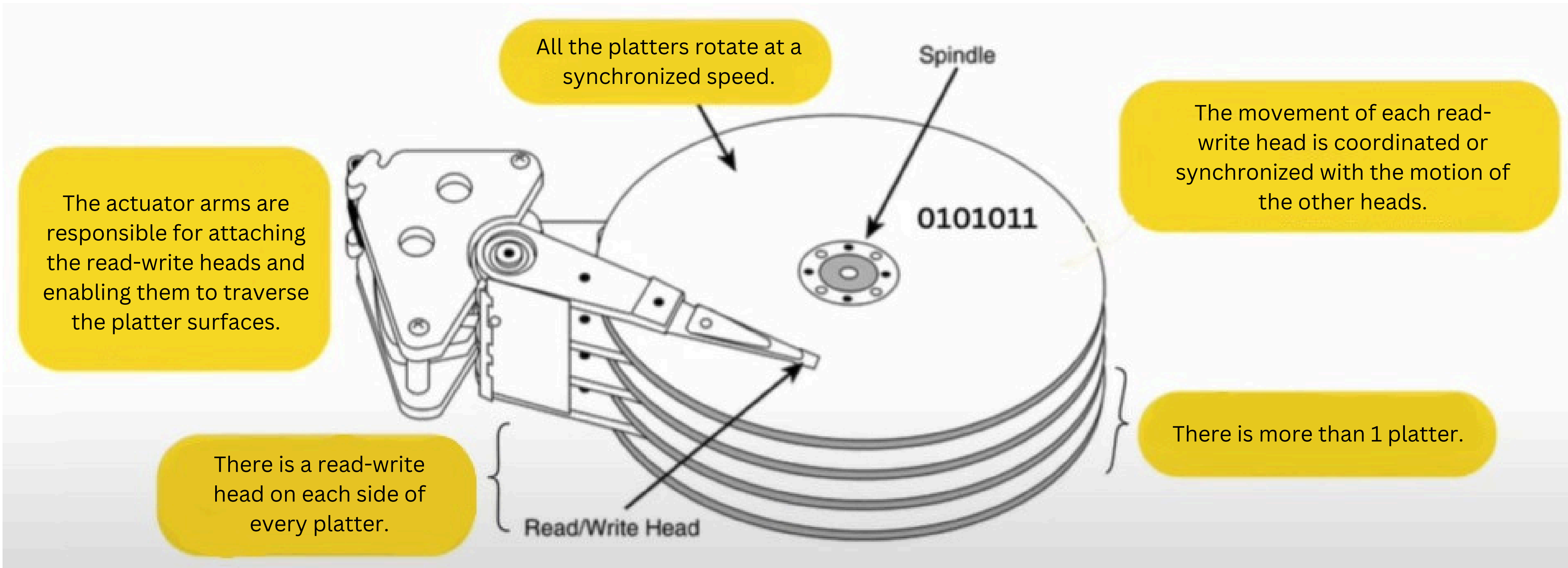
3.4 Secondary storage device

Magnetic Media

How?



3.4 Secondary storage device



3.4 Secondary storage device

Single Disk

Track

Circular paths on the HDD's surface used to organise data.

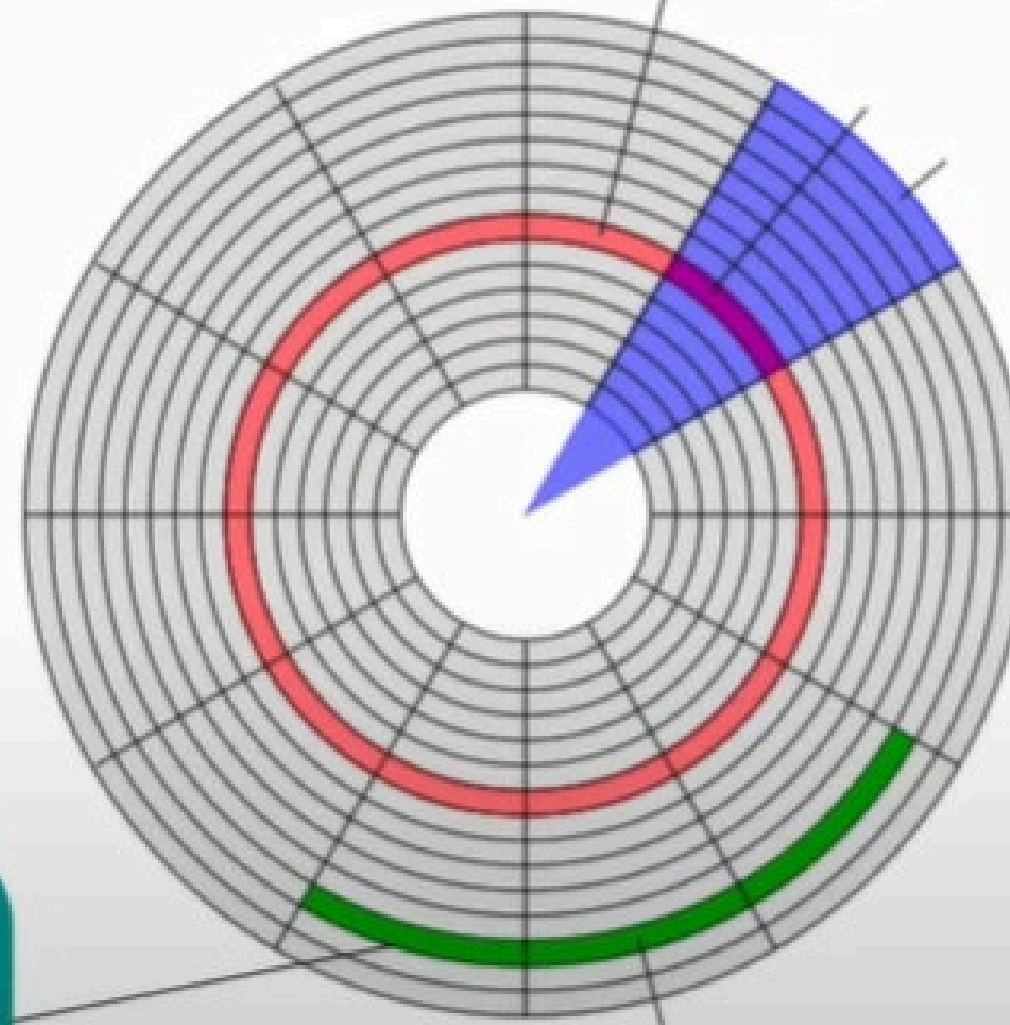
Sector

Small storage units within tracks, typically 512 bytes.

Due to the synchronized motion of the heads, it's possible for related data to be stored on the same tracks across different disks.

Block

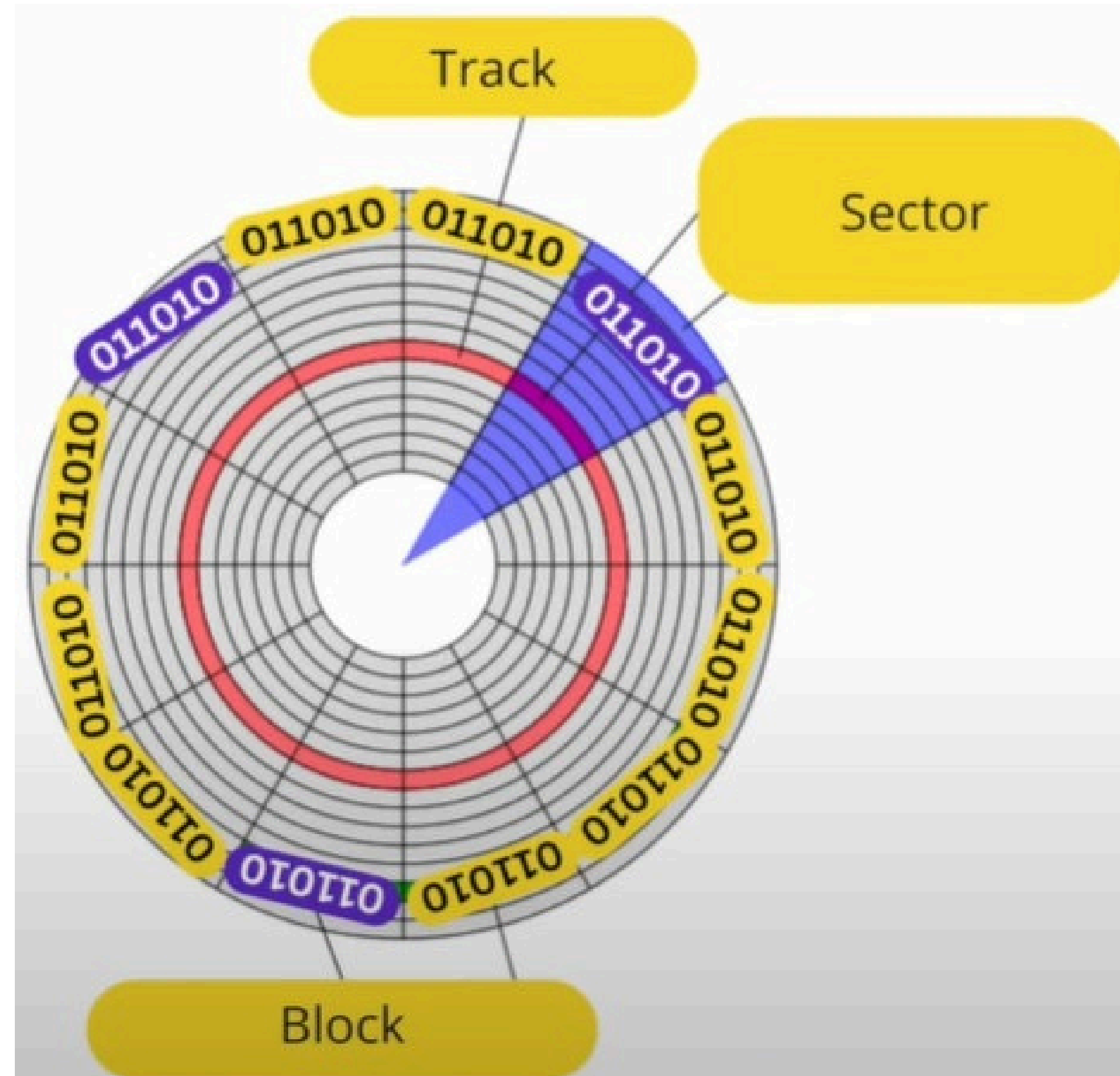
Groups of consecutive sectors used for file storage and management.



3.4 Secondary storage device

Defragmentation

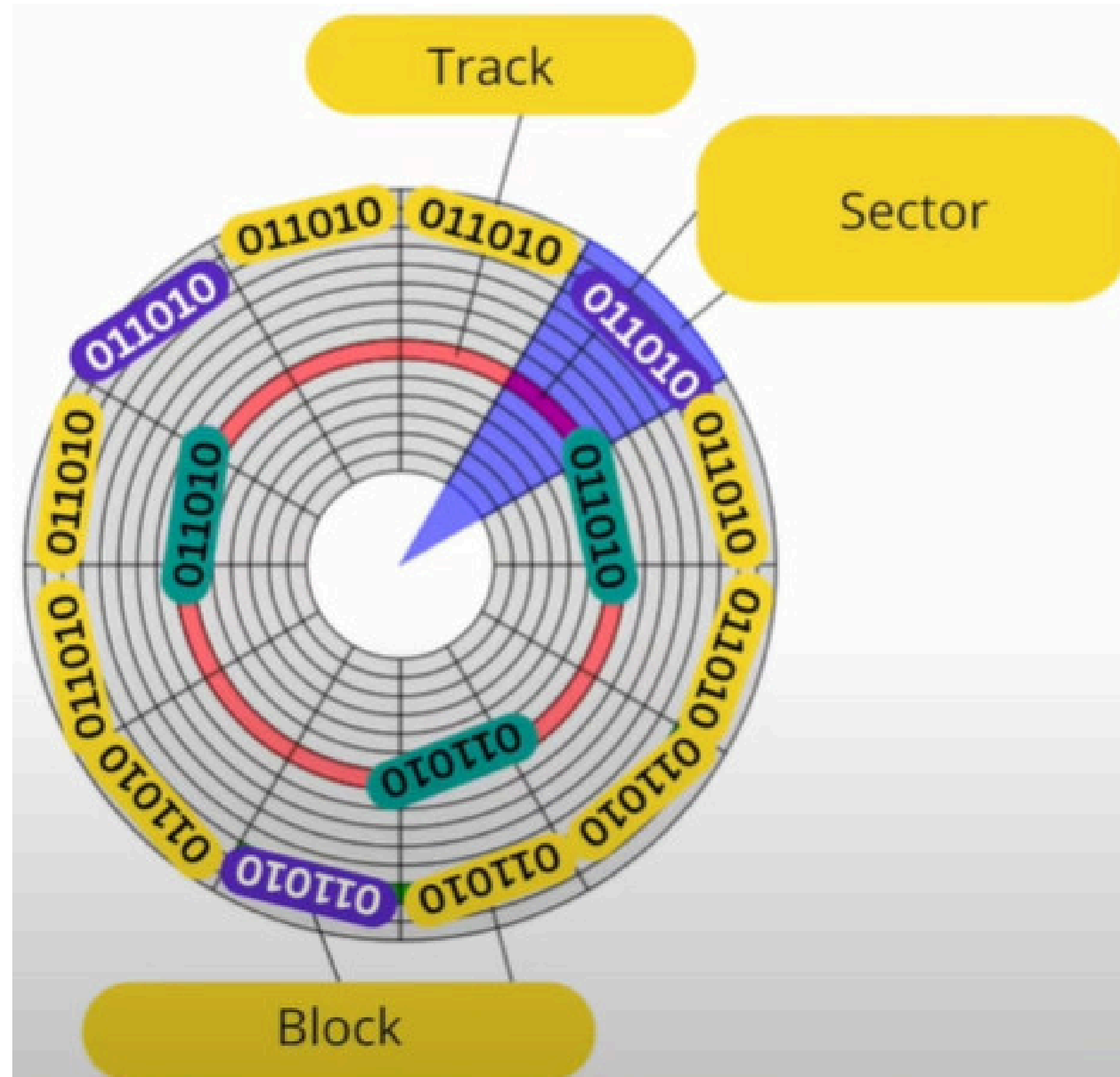
When storing a file, you need to allocate a necessary number of sectors, but they may not necessarily be contiguous (next to each other).



3.4 Secondary storage device

Defragmentation

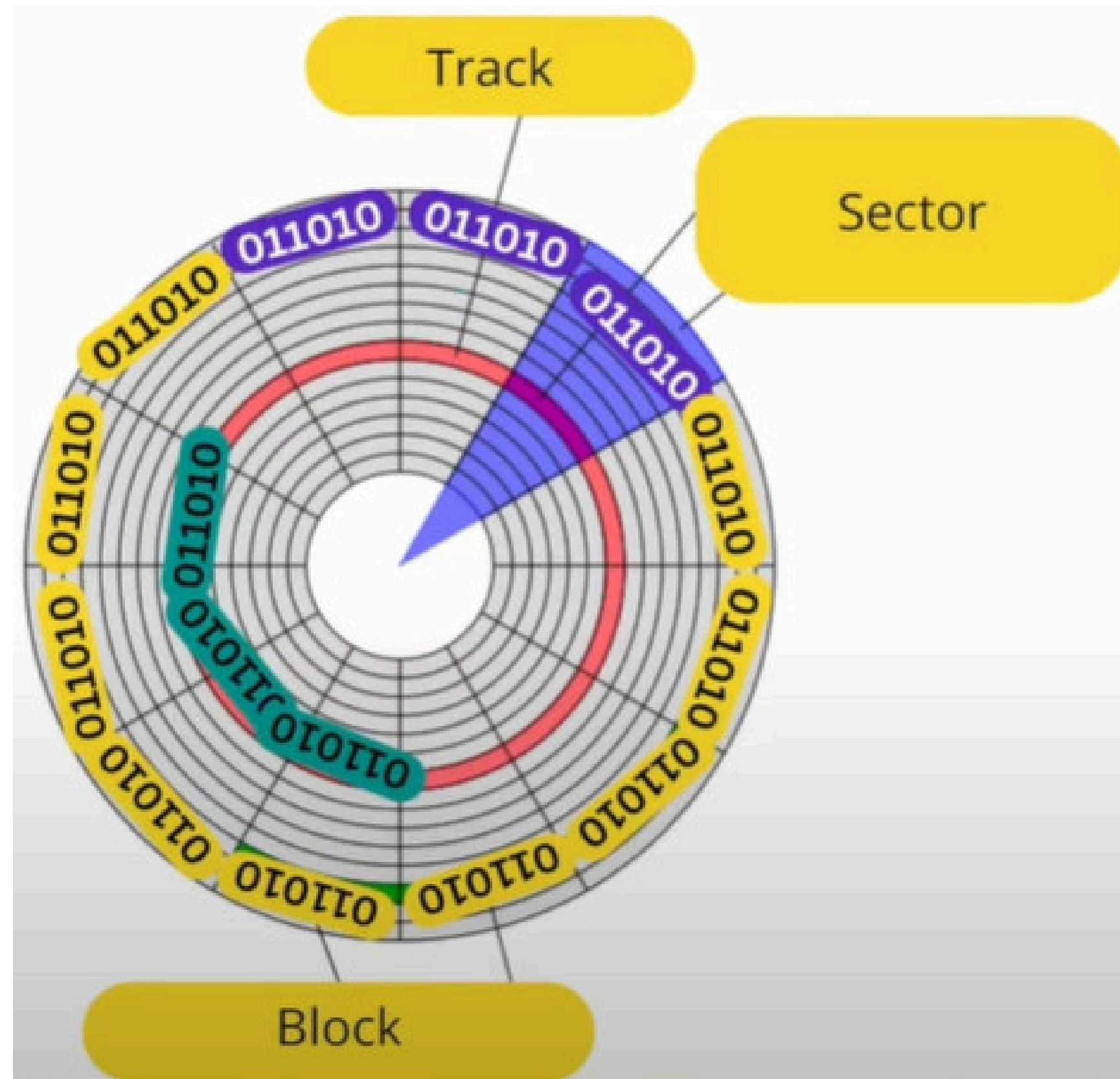
As files are created, deleted, or edited, the utilisation of sectors becomes more fragmented over time, leading to a degradation in disk performance.



3.4 Secondary storage device

Defragmentation

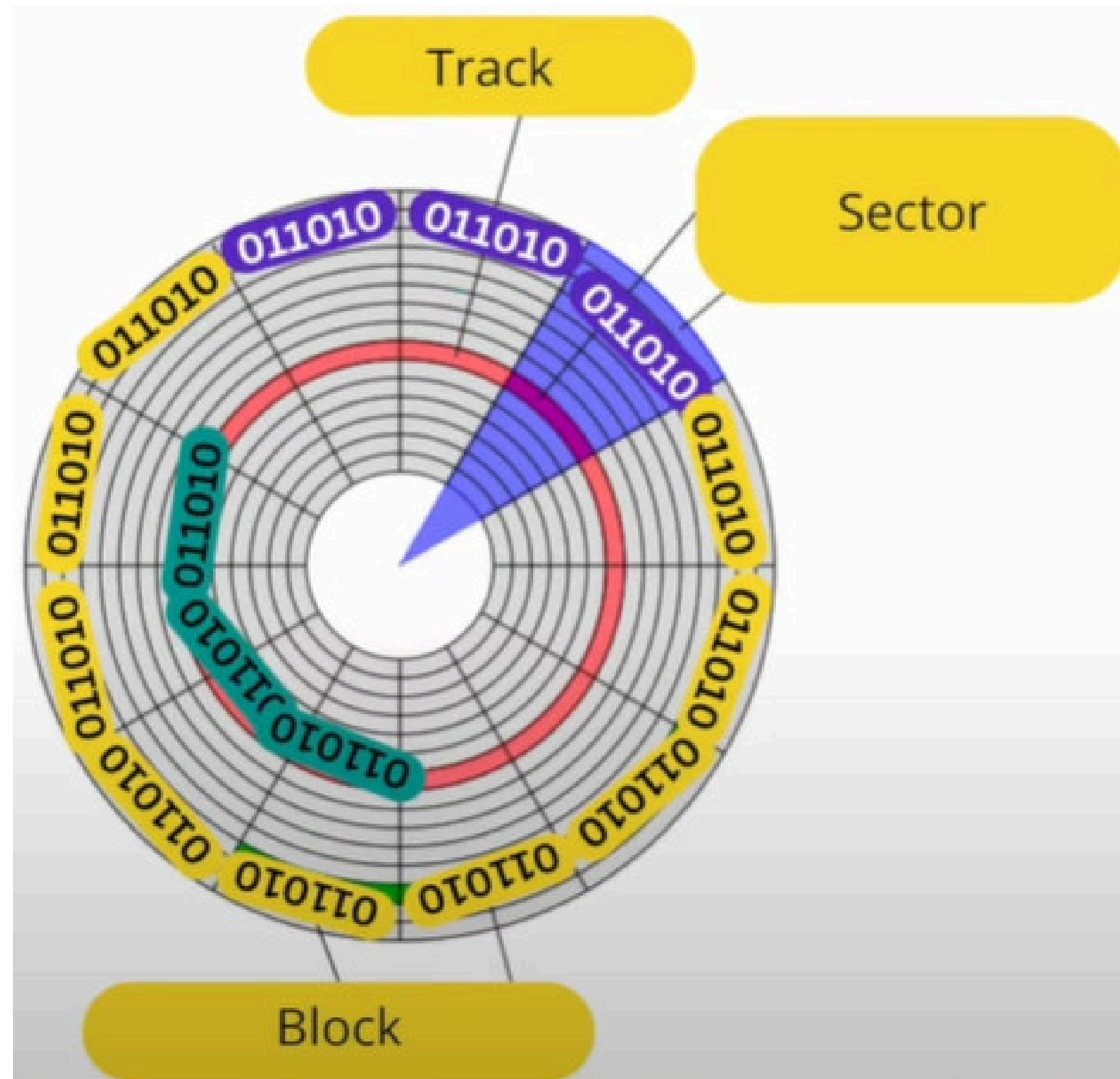
A defragmentation program can rearrange how sectors are allocated to files, restoring disk performance.



3.4 Secondary storage device

Hard Disk

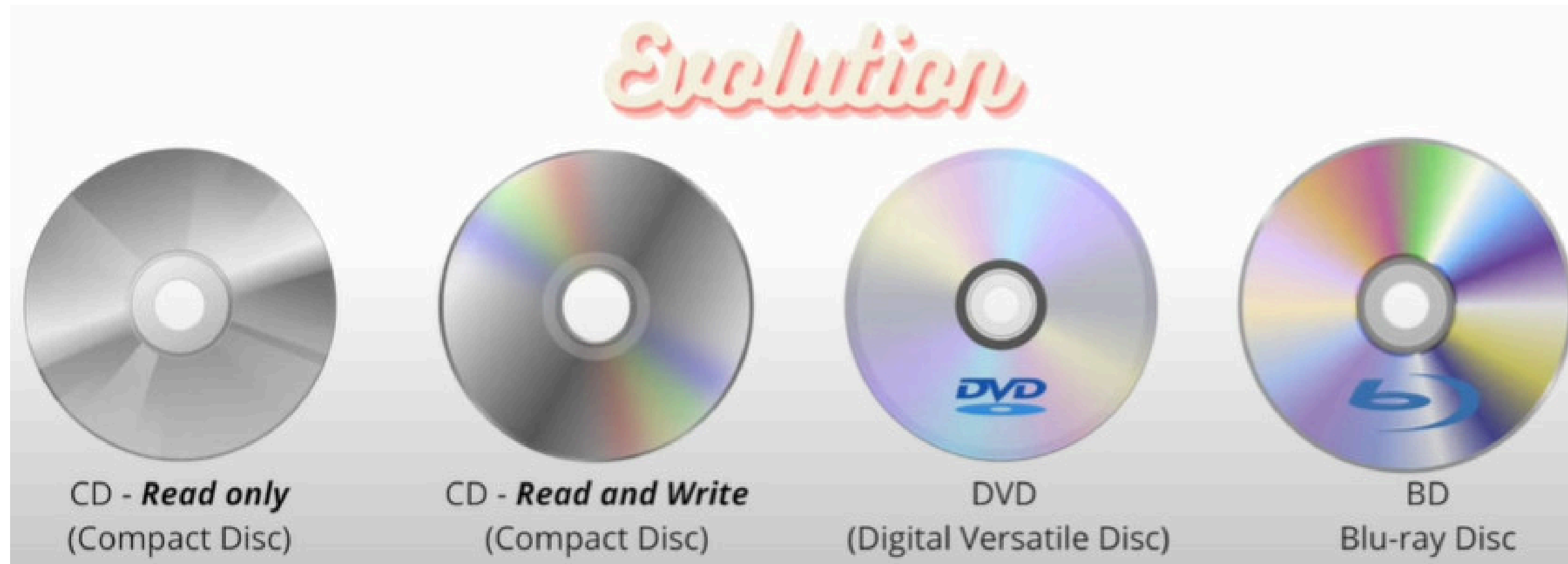
A hard disk is classified as a direct-access read-write device since any sector can be selected for reading or writing. However, it's important to note that the data within a sector must be read sequentially.



3.4 Secondary storage device

Optical Media

Optical media is a storage medium that uses laser technology to read and write data on discs, including formats like CDs, DVDs, and Blu-ray discs.



3.4 Secondary storage device

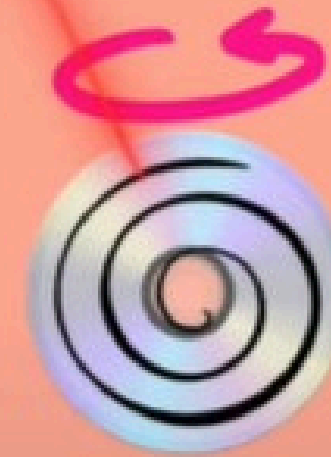
How it works

The optical disc features a single continuous spiral pathway that extends from the innermost part of the surface to the outer periphery.



*Simplified version

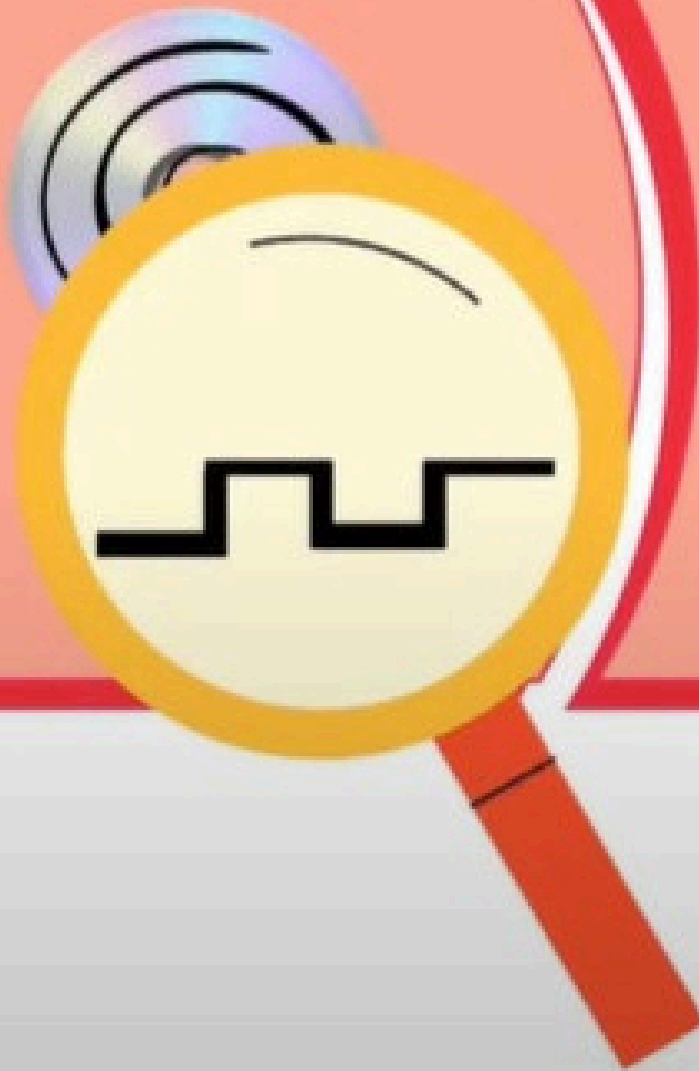
During operation, the disc spins. Simultaneously the laser moves across ensuring that it is continuously focused on the spiral track.



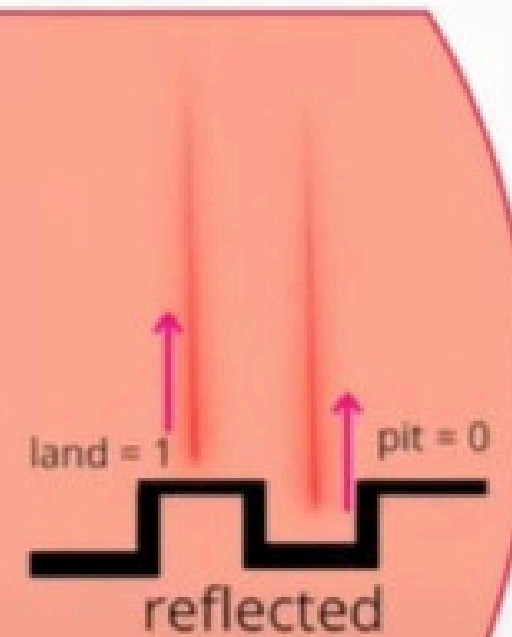
3.4 Secondary storage device

How it works

The surface of the disc contains features known as pits and lands within the track.



The laser beam bounces off the disc's surface through reflection. Pits are interpreted as binary 0, while lands (flat areas) are interpreted as binary 1 in the digital encoding of data on optical discs like CDs.



3.4 Secondary storage device

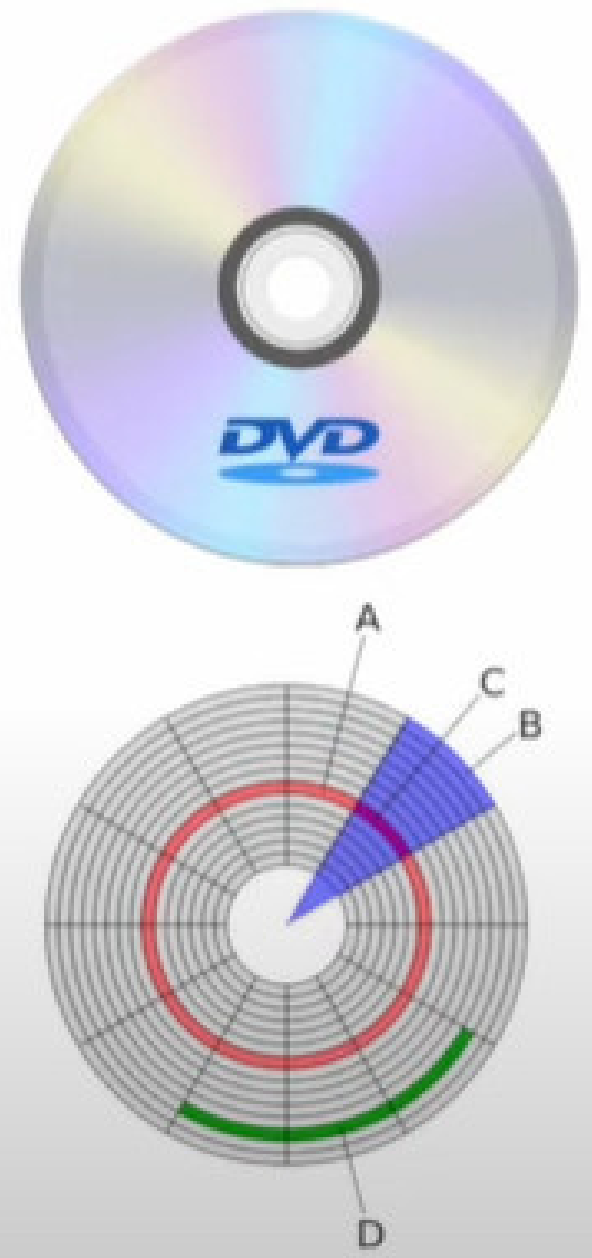
CD-RW, DVD-RW



- CD-RW and DVD-RW use a unique alloy material as their reflective surface.
- During the writing process (burning), laser heat turns the material into a liquid state.
- The intensity of the laser determines whether the material becomes crystalline or amorphous when it cools.
- During reading, laser light reflects from crystalline solids but not amorphous ones, enabling the encoding of 1s and 0s.

3.4 Secondary storage device

Property



- The disc, despite having a single track, functions as a direct-access device. The disc, despite having a single track, functions as a direct-access device.
- This is possible because the laser can move both forward and backward along the track.
- Data on the disc is organized into sectors, similar to how data is formatted on a magnetic hard disk.

3.4 Secondary storage device

Changing Storage Capacity

Two ways

- Speed of rotation
- Wavelength of the light; Blue ray has shorter wavelength, allow more pits and lands to be gathered per unit area. Hence allowing more storage capacity.



The diagram illustrates the difference in pit patterns between a DVD and a Blu-ray disc. On the left, a DVD is shown with a pit pattern of 400x320 nm and a diameter of Ø1.1 µm. On the right, a Blu-ray disc is shown with a pit pattern of 150x130 nm and a diameter of Ø0.48 µm. The Blu-ray disc has a much denser pattern of smaller pits, which allows for higher storage capacity.

3.4 Secondary storage device

Solid State Media

It is a semiconductor technology that has no moving parts.

	000		000
	001		001
	010		010
	011		011
	100		100
Electrons	101	Electrons	011
Electrons	110	Electrons	100
Electrons	111	Electrons	101
		Electrons	110
		Electrons	111

The circuits comprise arrays of transistors functioning as memory cells.

NAND is the most commonly utilized technology due to its basic circuitry's similarity to that of a NAND logic gate.

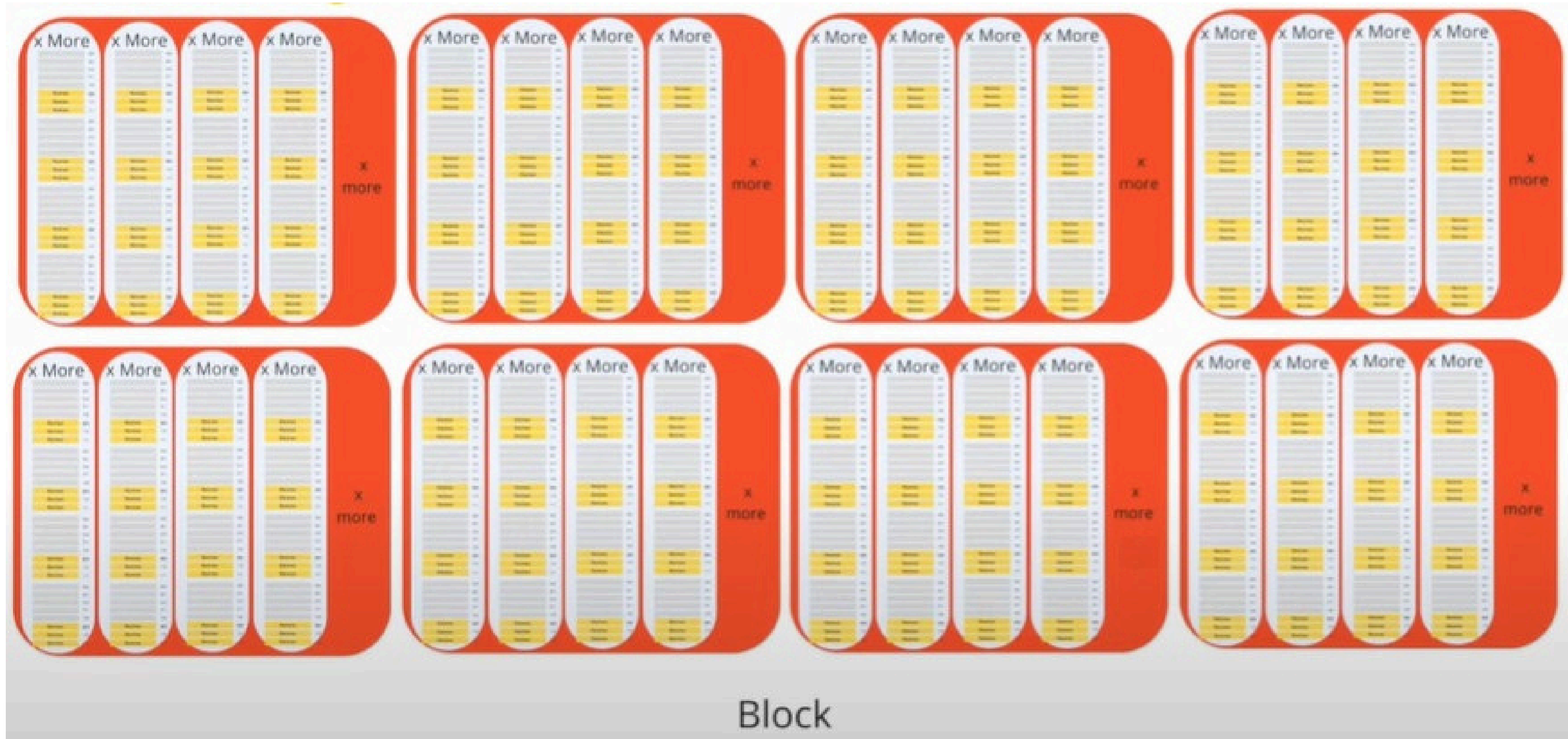
3.4 Secondary storage device

Terms



3.4 Secondary storage device

Terms



3.4 Secondary storage device

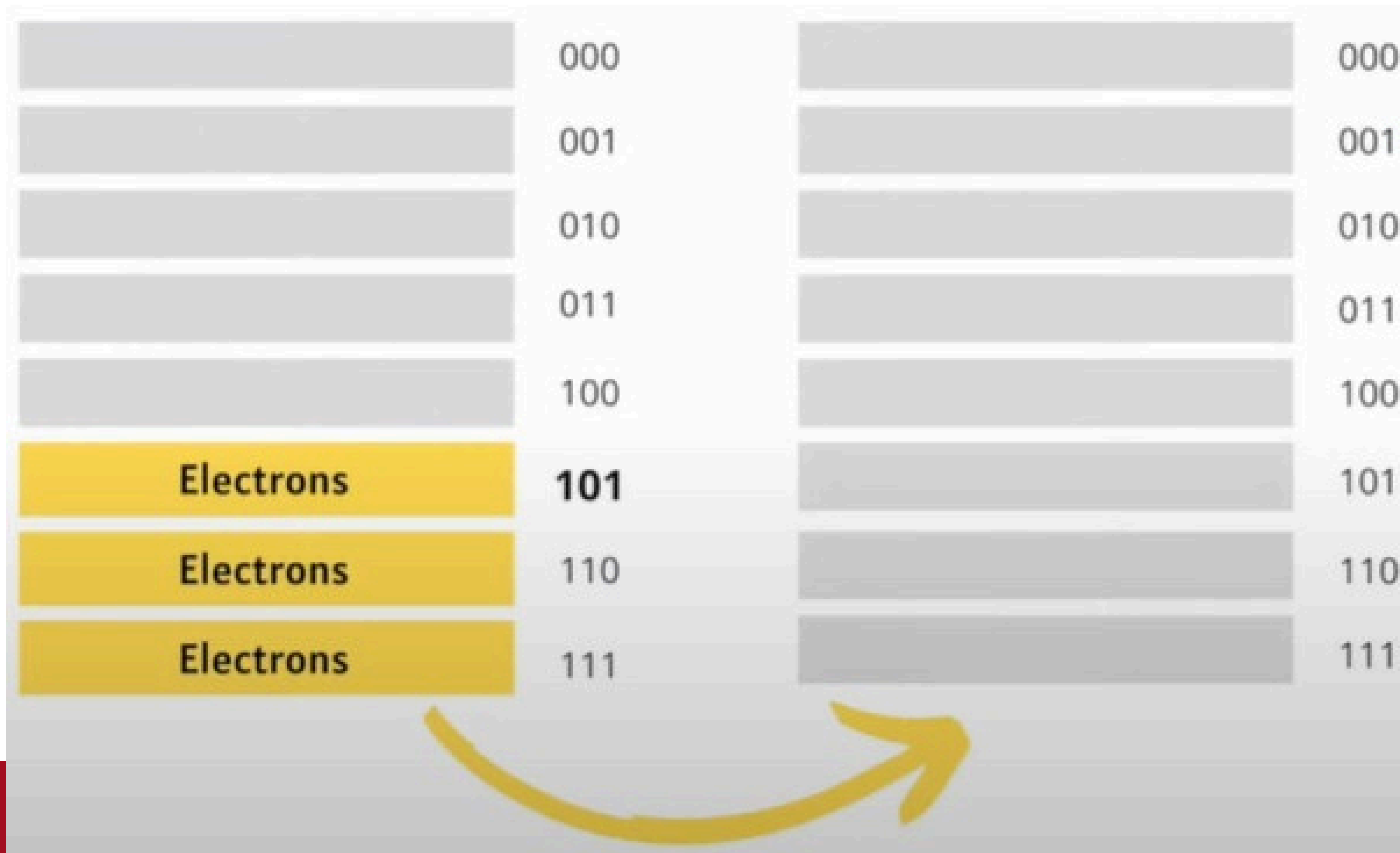
Terms



3.4 Secondary storage device

Solid State Media

Before writing data to a memory block, the block's existing data must be erased. This unique capability allows for the erasure of an entire block of memory cells in an instant.



3.4 Secondary storage device

Solid State Media

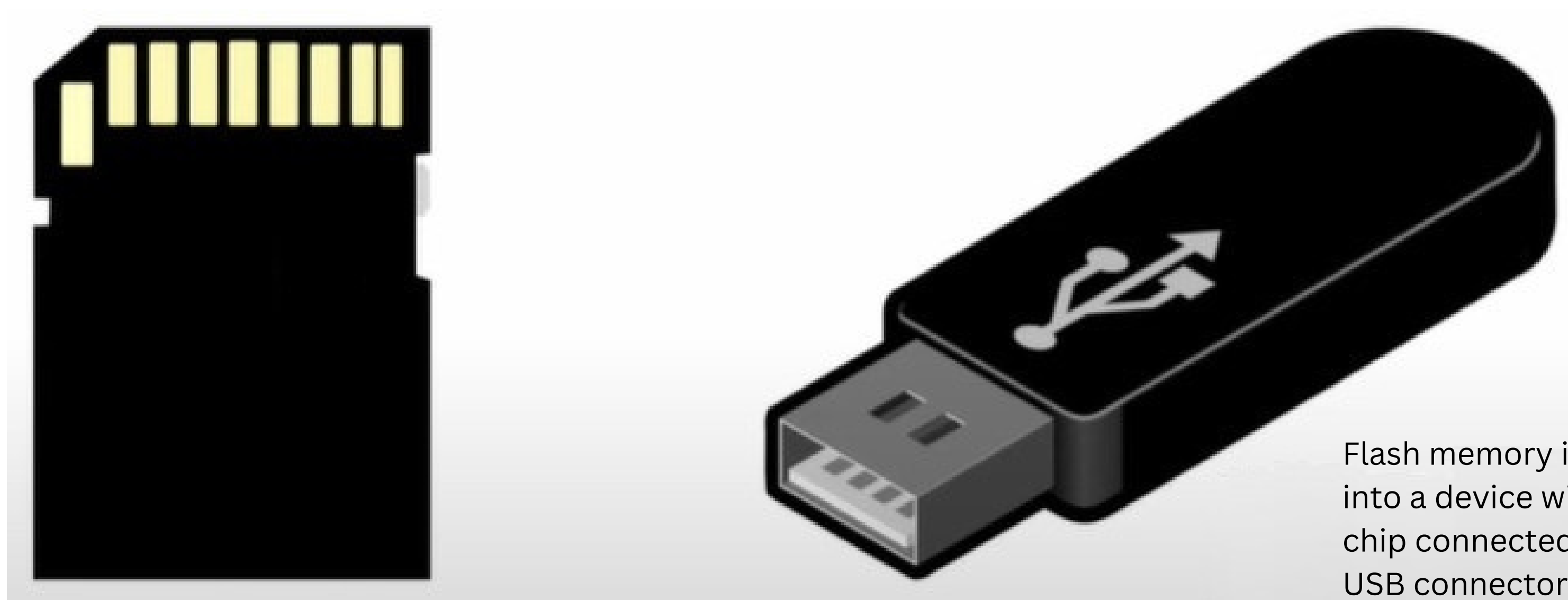
When data is read, a single page of data can be read in one operation.



3.4 Secondary storage device

Use cases

The most common use for solid-state media is either in a memory card or a USB flash drive.



Flash memory is integrated into a device with the memory chip connected to a standard USB connector.

3.4 Secondary storage device

Solid State Drive

A solid-state drive (SSD) is a storage device that uses NAND-based flash memory to store data, offering faster performance and durability compared to traditional hard disk drives (HDDs).



While degradation is possible, it will occur gradually and can be identified and mitigated through corrective measures.

3.4 Secondary storage device

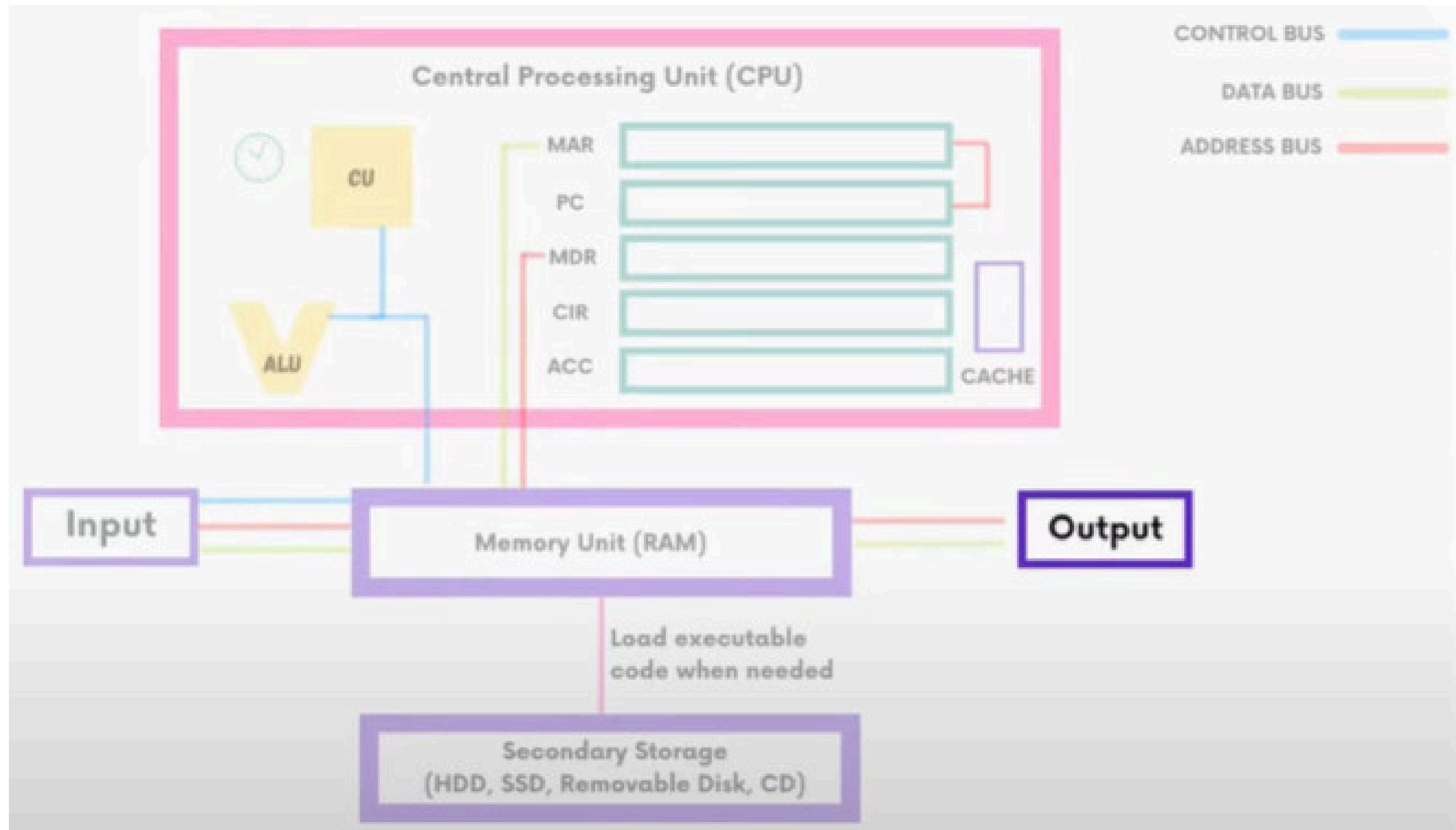
Solid State Drive



SSD is

- Lighte
- More Reliable
- Thinner
- Faster in data access
- Cooler

3.5 Output devices



3.5 Output devices

Screen Display

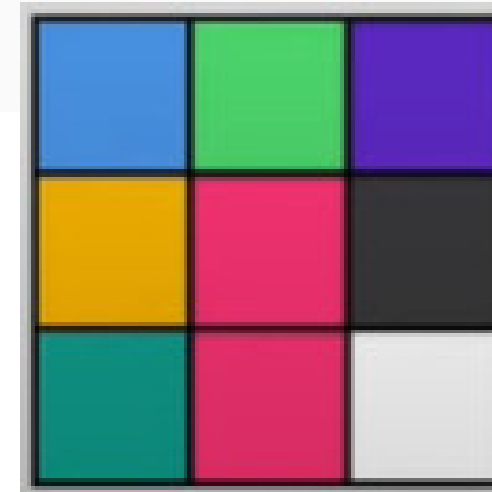


11011010

10110111

01101111

A screen pixel comprises three sub-pixels, typically representing red, green, and blue colours. Adjusting the brightness of these individual sub-pixels enables the display of a wide spectrum of colors.

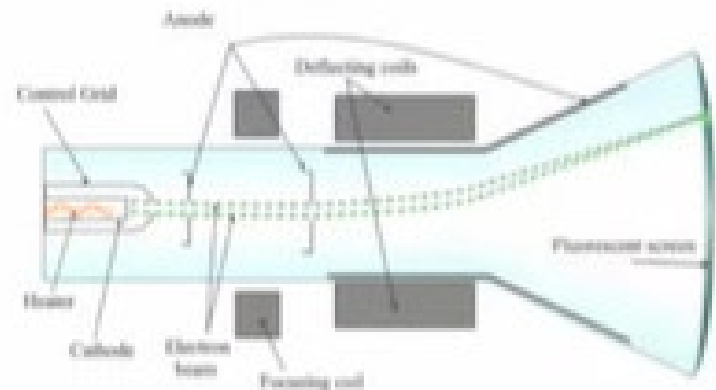


An image can be stored as a bitmap created from individual pixels.

3.5 Output devices

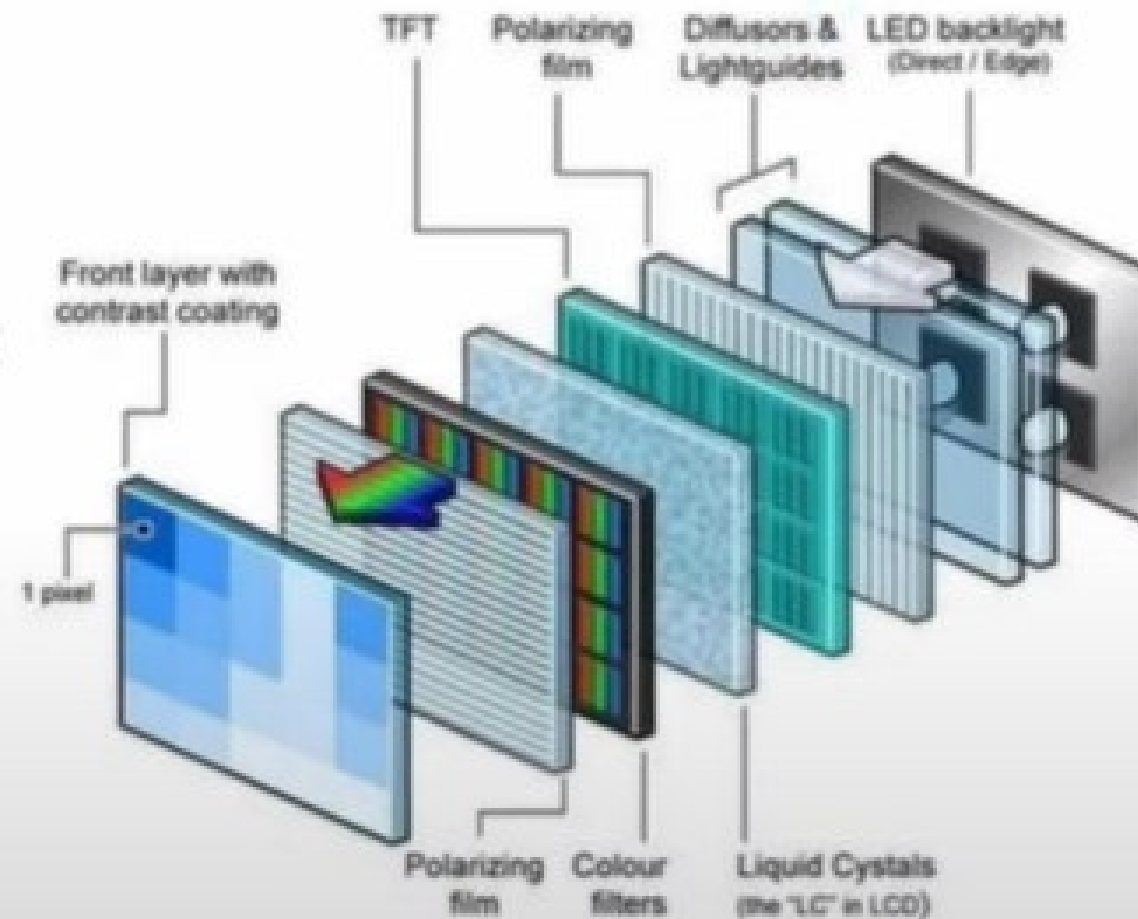
Evolution

Cathode Ray Tube (CRT)



- A cathode ray tube (CRT) uses an electron gun to emit a stream of electrons.
- These electrons are directed and controlled by electromagnets to strike a phosphorescent screen, producing light and creating the images you see on the screen.

Liquid Crystal Display (LCD)



- Each pixel contains a liquid crystal cell.
- LED backlighting is used to illuminate the pixel matrix.
- Polarized light is directed at the pixel matrix.
- A polarizer is placed between the pixel matrix and the screen.
- When voltage is applied to a pixel cell, it changes the liquid crystal molecule alignment.
- This change in alignment alters the light's polarization and the displayed screen content.

3.5 Output devices

Virtual Reality Headset



- A virtual reality headset is a **wearable** device that immerses the user in a computer-generated, **three-dimensional environment**, often incorporating head and body tracking for an interactive and immersive experience.

3.5 Output devices

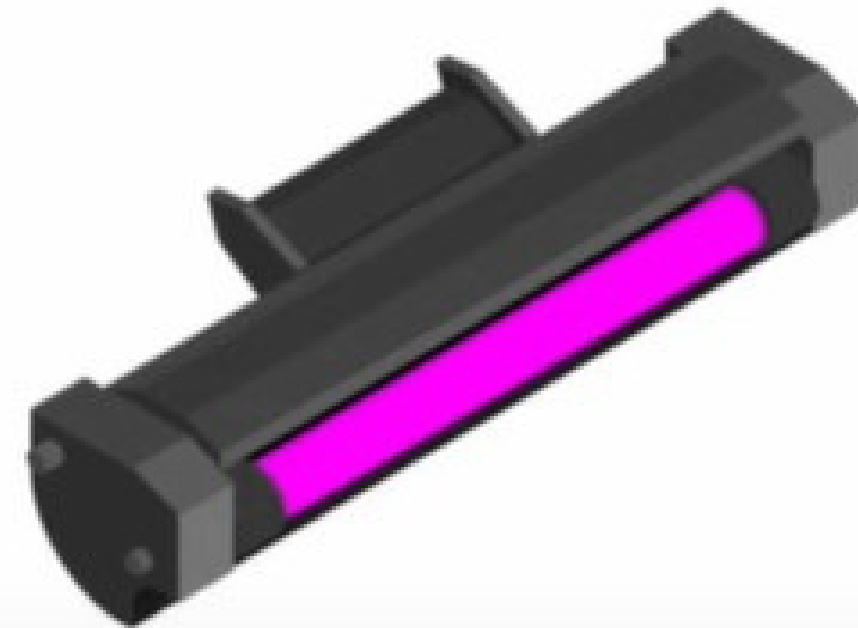
Hard-copy output of text

Inkjet Printer



A type of printer that sprays tiny droplets of ink onto paper to create text and images.

Laser Printer



A type of printer that uses a laser beam to create an electrostatic image on a photosensitive drum, which is then transferred and fused onto paper to produce text and images.

3.5 Output devices

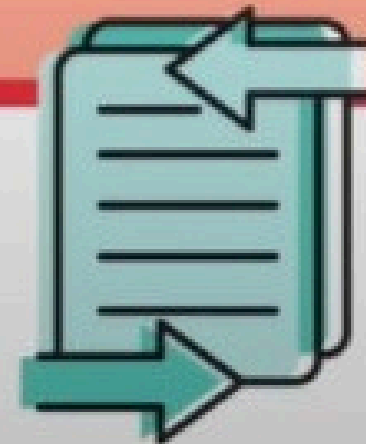
How it works Inkjet Printer



A sheet of paper is inserted, and subsequently, the printhead traverses across the sheet, applying ink onto the paper.

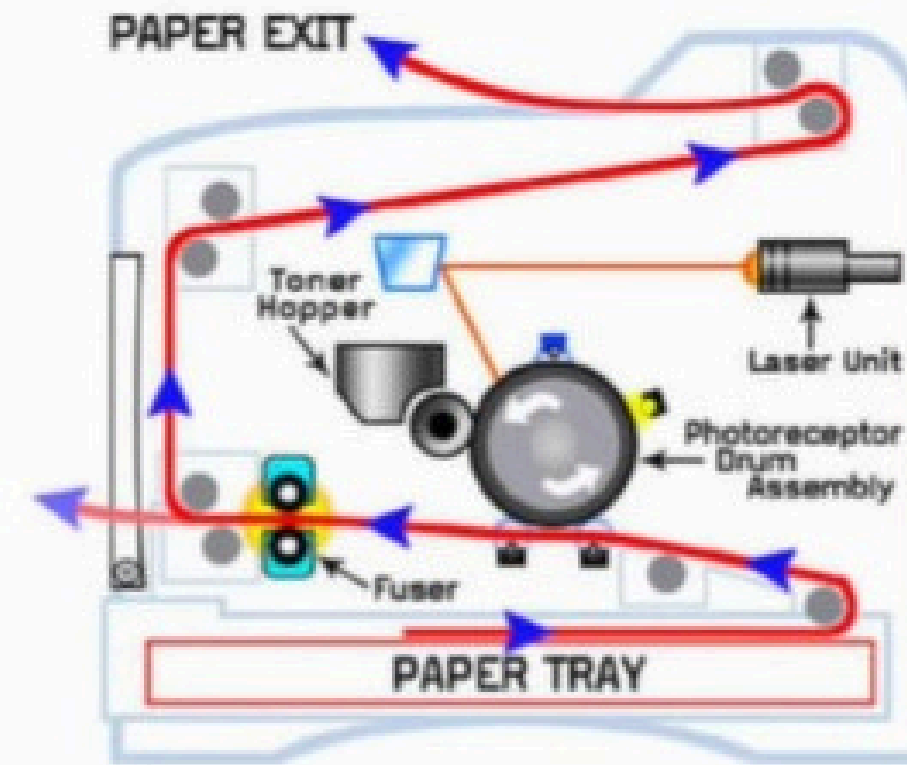
The paper is advanced slightly, and the printhead repeats its movement across the paper, with this process continuing until the entire sheet has been completely printed.

The printhead is composed of nozzles that eject droplets onto the paper, and ink is provided to the printhead from one or multiple ink cartridges.



3.5 Output devices

How it works Laser Printer



The drum is electrically charged. The revolving drum is exposed to a laser beam guided by mirrors and lenses, selectively discharging its surface at various positions.

This repetition creates a full-page electrostatic image. Charged toner adheres only to the discharged areas on the drum's surface.

The toner is transferred to a charged sheet of paper, fused using heated rollers, and the drum is discharged before repeating the process for the next page.

3.5 Output devices

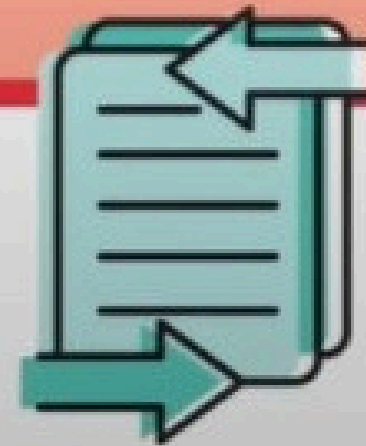
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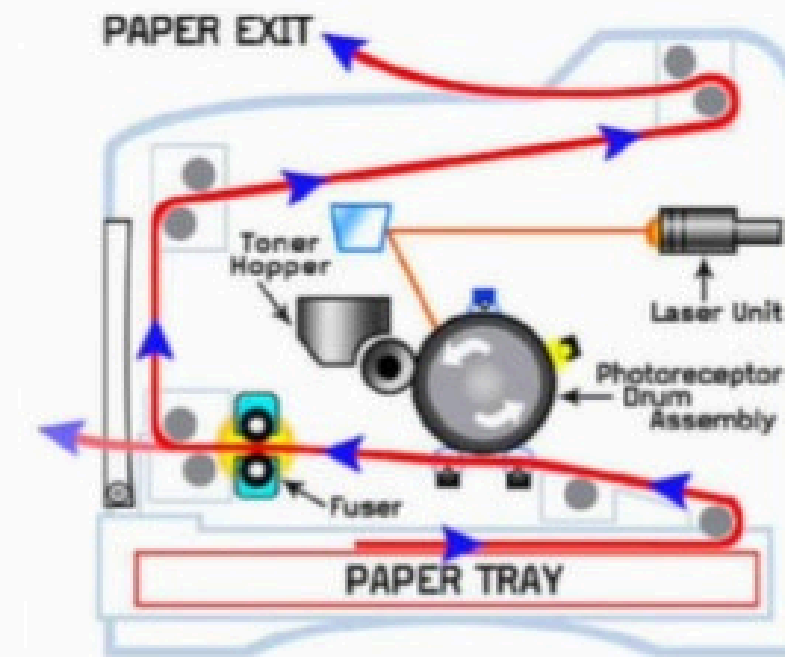
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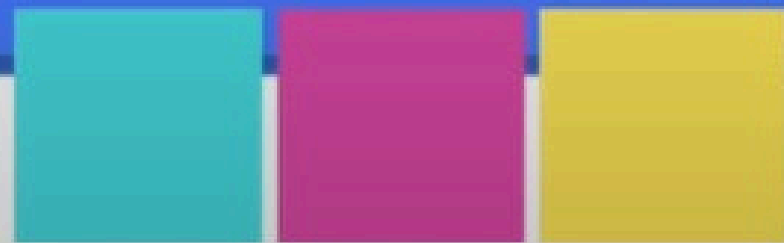


3.5 Output devices

How it works Laser Printer



Color printing necessitates separate toners for cyan, magenta, yellow, and black, with each color processed individually.

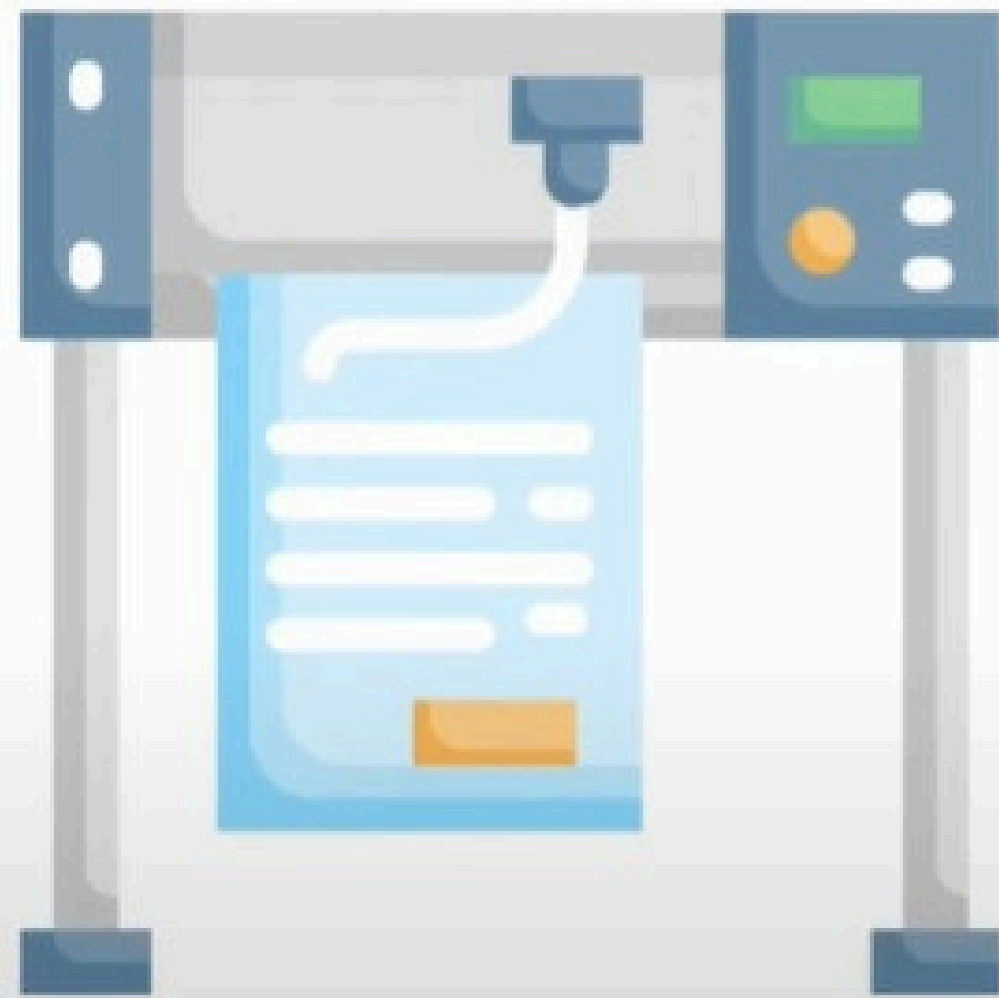


The technology generates dots, and image quality is influenced by the dots per inch, which can be controlled by software to adjust the number of dots per pixel.

3.5 Output devices

Hard-copy graphic output

Graphic Plotter



Graphic Plotter: A graphic plotter is a specialized device for drawing precise, detailed illustrations and designs on paper.

Pen-based Drawing: It works by using pens or drawing tools to move across the paper, creating lines and shapes according to digital instructions.

Vector Graphics: Graphic plotters excel at producing vector graphics with consistent line quality, making them valuable in engineering and design application.

3.5 Output devices

Hard-copy graphic output

3D Printer

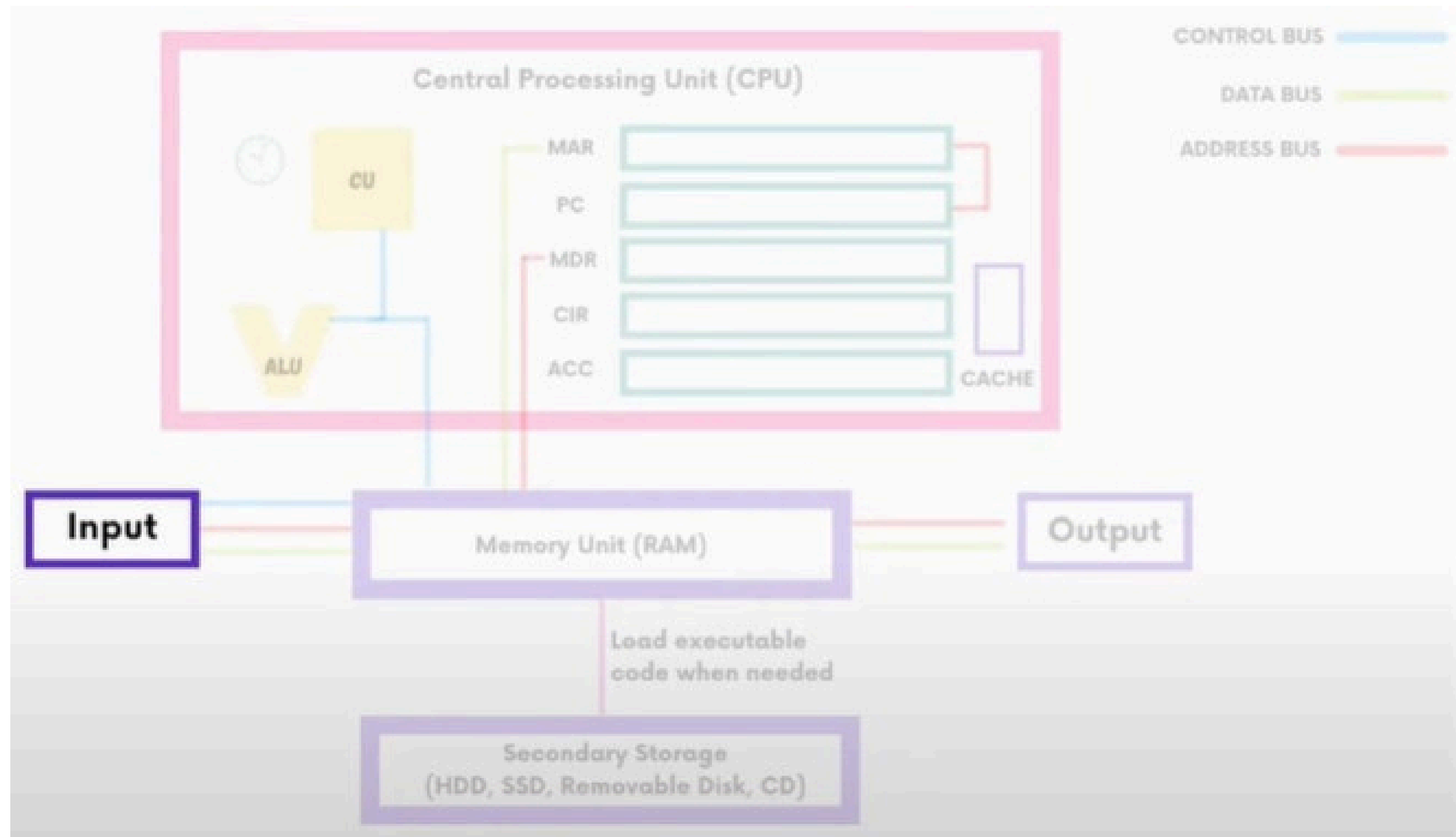


● 3D Printer: A 3D printer is a machine that creates physical objects layer by layer from digital 3D models.

● Mechanism: The design is split into layers. Each layer is sent to the printer. The 3D printer uses a nozzle to squirt material onto the printer bed to create a physical layer. This is repeated for many layers. The object will then be cured to ensure all layers stuck together.

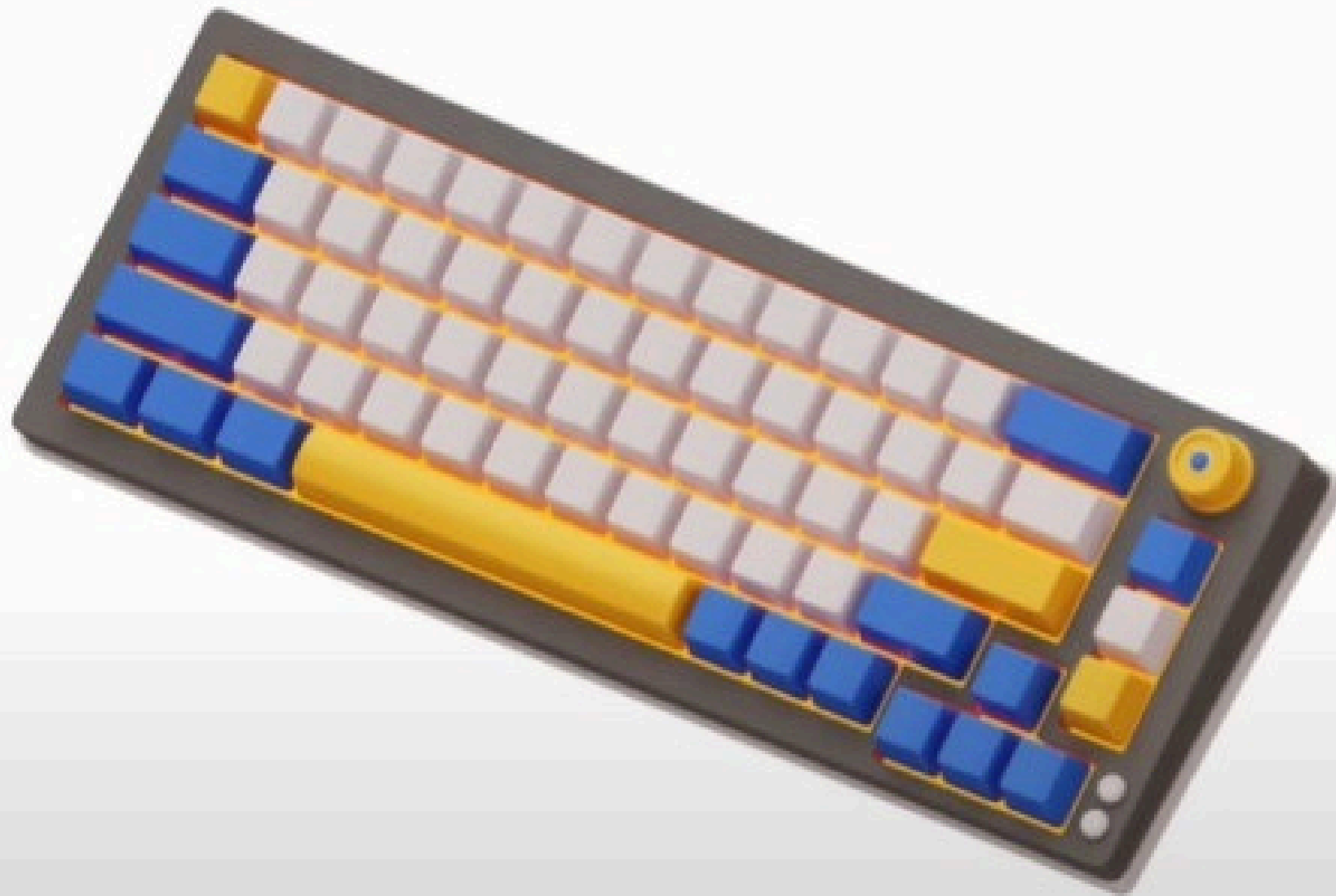
● Diverse Applications: 3D printers are used in various fields, from prototyping and product design to healthcare and aerospace for creating custom parts and prototypes.

3.6 Input device



3.6 Input device

Keyboard

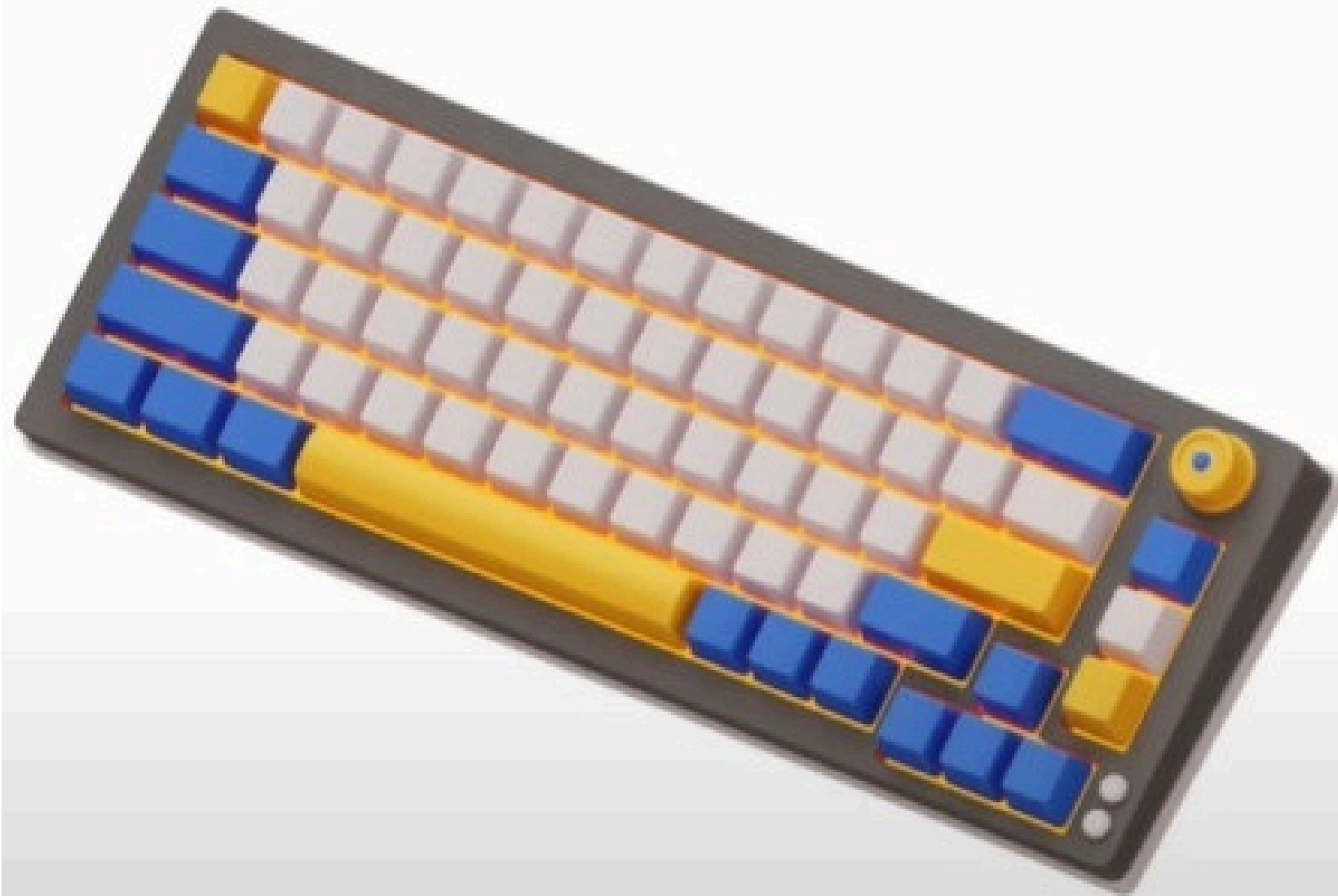


Functionality

- The keyboard allows users to input text data, creating an illusion of immediate character transfer to the computer screen when a key is pressed.
- In reality, key presses are converted into **character codes**, which are then transmitted to the processor.
- The **operating system**, under its control, ensures that the text character appears on the screen.
- The same process occurs when using keyboard shortcuts to initiate actions, with the processor responding by executing the requested action.

3.6 Input device

Keyboard

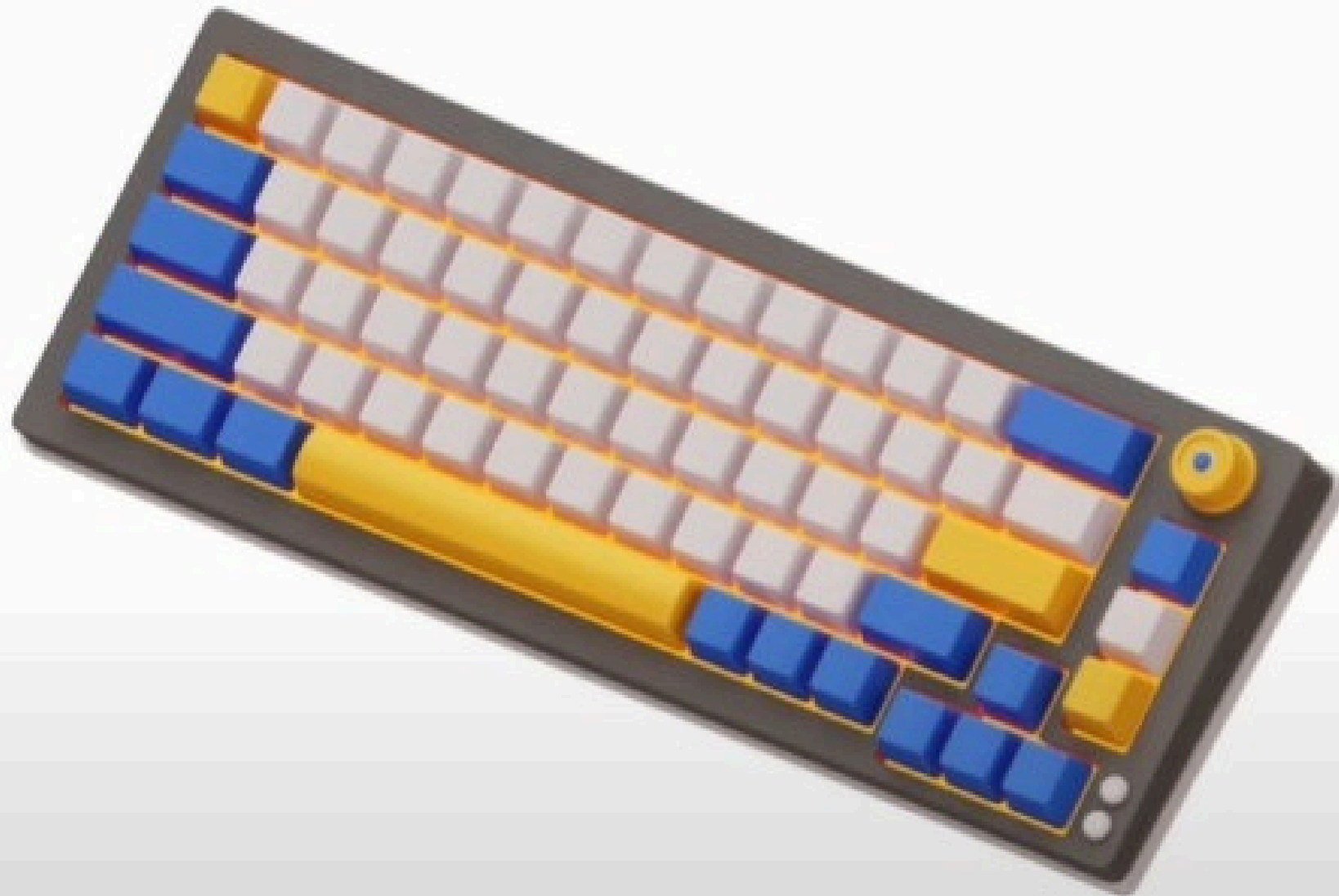


Keyboard Components

- The keyboard incorporates electrical circuitry, along with its own microprocessor and a ROM (Read-Only Memory) chip.

3.6 Input device

Keyboard



Key Recognition Process

- The keys are positioned above a key matrix, consisting of rows and columns of wires.
- Pressing a key establishes contact at the intersection where wires cross.
- A continuous test is conducted by the microprocessor to identify closed electrical circuits between row and column wires.
- When a closed circuit is detected, the microprocessor identifies the specific intersection responsible.
- Data stored in the ROM is then used to determine the character code associated with the pressed key.
- This character code is subsequently sent to the screen for display.

3.6 Input device

Screen

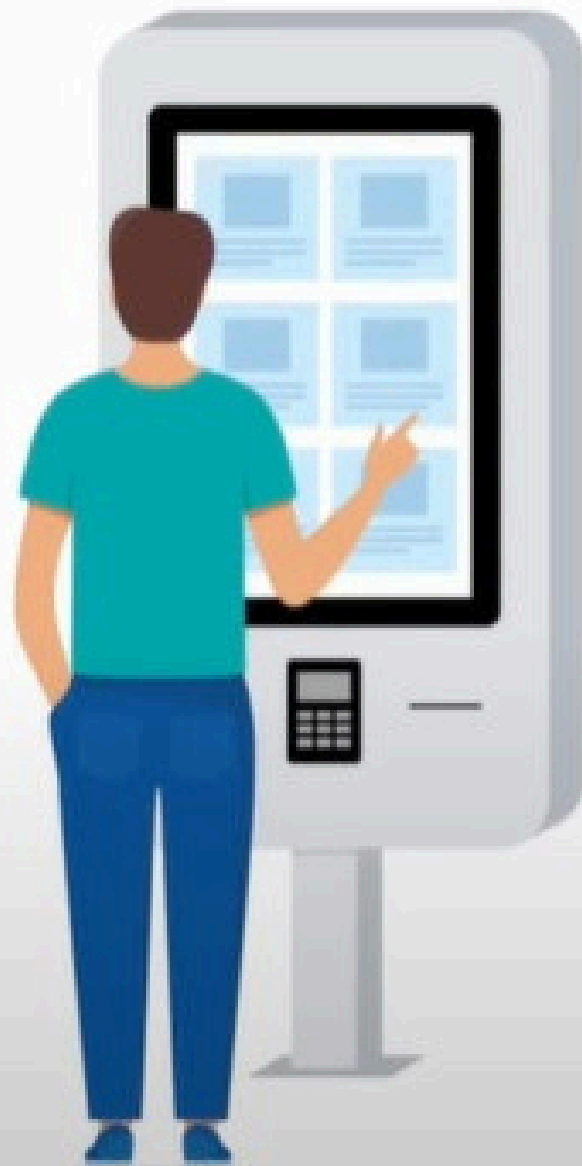


Graphical User Interface (GUI)

- GUI allows the user to control data input by providing different types of screen icon.
- Pointing device (eg. A mouse) is required.

Touch Screens

Resistive Touch Screens

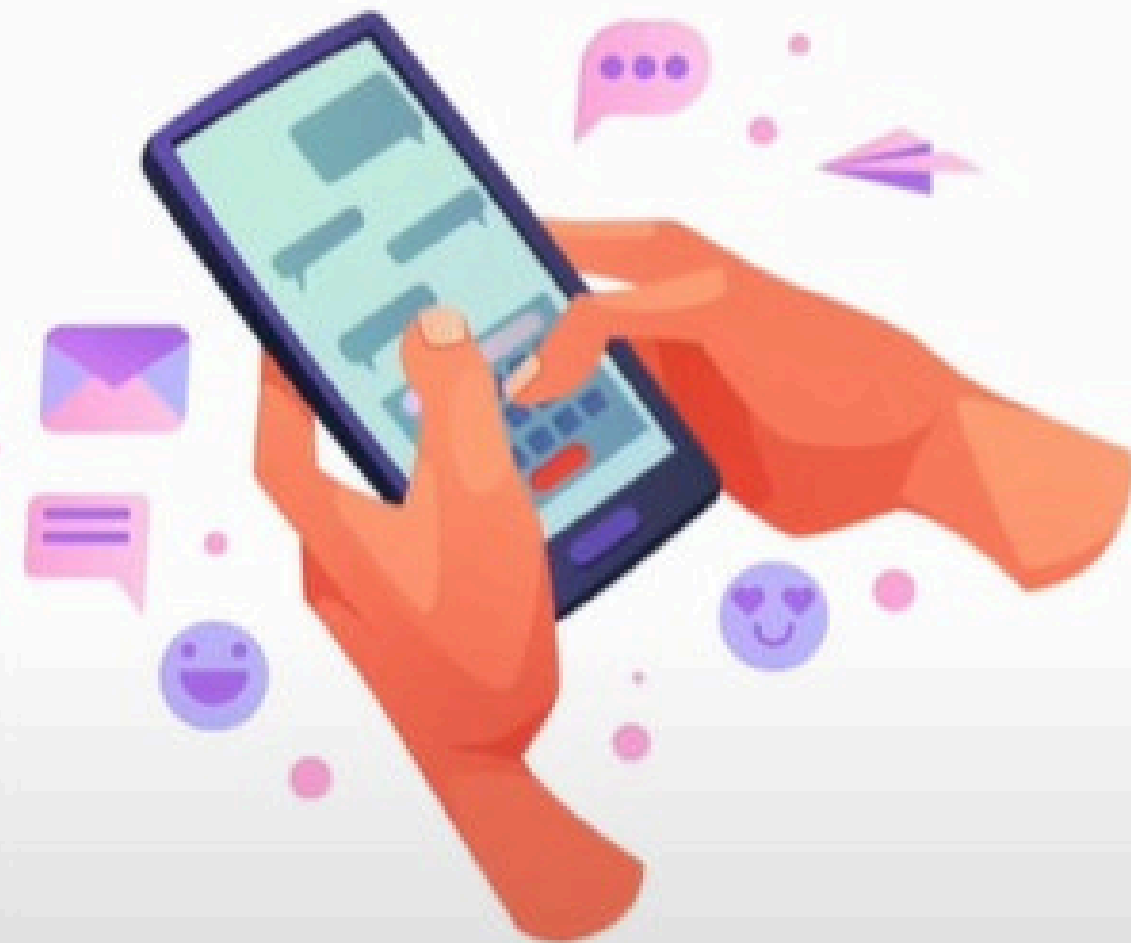


- Resistive touch screen consists of two flexible layers separated by a thin space.

- When pressure is applied, the top layer moves to make contact with the lower layer, creating voltage dividers in both horizontal and vertical directions.

Touch Screens

Capacitive Touch Screens



- Capacitive touch screens operate by detecting capacitance changes caused by finger touches on a glass screen.

- Projective capacitive touch (PCT) technology uses mutual capacitance and features a circuit with an array of capacitors under the screen, enabling multitouch detection and versatile applications.

3.6 Input device

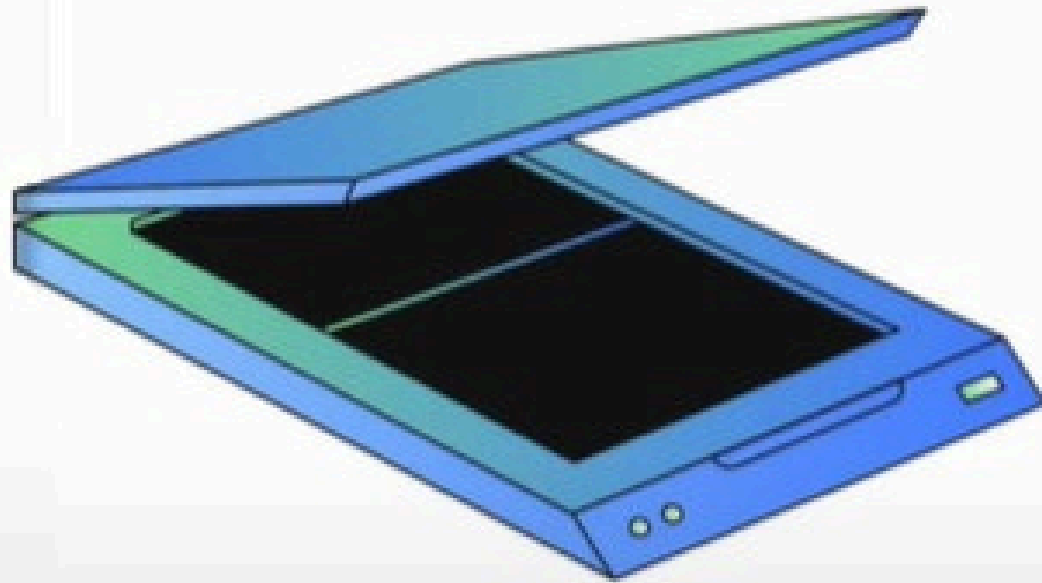
Input of a graphic



● A webcam serves as a tool for transmitting video footage to a computer system, while a digital camera can be linked to a computer to transfer and save photos or videos within the computer's storage.

3.6 Input device

Input of a graphic



A scanner operates in contrast to the printing process by converting an image into a digital form. To do this, it keeps a sheet of paper with the image stationary while a light source moves across its width, spanning from one edge to the other. The reflected light is then guided by a system comprised of mirrors and lenses onto a charge-coupled device (CCD).

CHARACTERISTICS OF A CCD

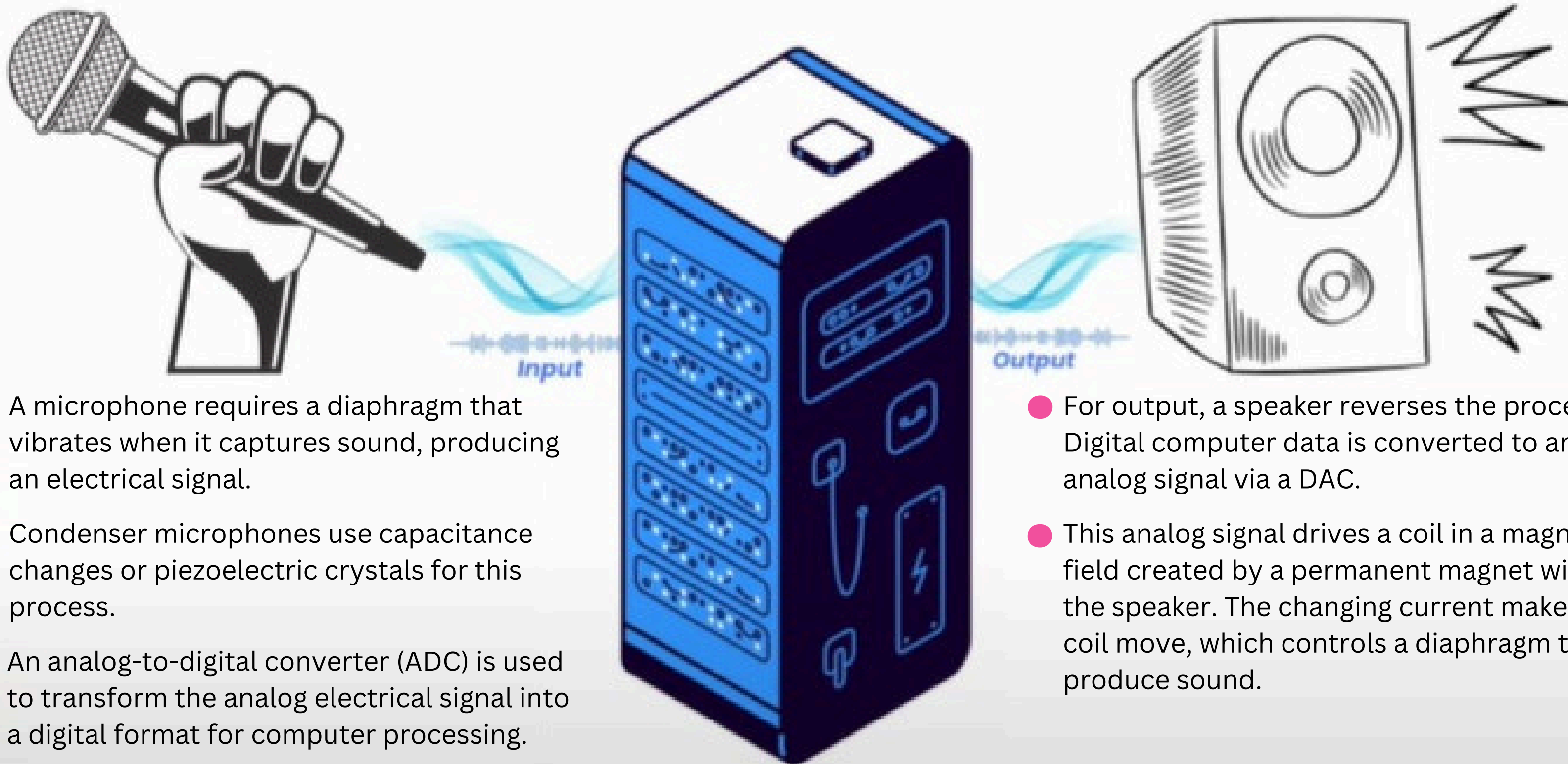
- A CCD comprises a grid of light-sensitive cells.

- Each cell within a CCD generates an electrical signal that corresponds to the intensity of light it receives.

- To facilitate communication with a computer, a CCD requires an analog-to-digital converter to transform its analog signals into digital values.

3.7 Input and output for sound

There are many applications that require both voice input and voice output:
Video conferencing, IP telephony, and voice assistant (Siri).



- A microphone requires a diaphragm that vibrates when it captures sound, producing an electrical signal.
- Condenser microphones use capacitance changes or piezoelectric crystals for this process.
- An analog-to-digital converter (ADC) is used to transform the analog electrical signal into a digital format for computer processing.

- For output, a speaker reverses the process. Digital computer data is converted to an analog signal via a DAC.
- This analog signal drives a coil in a magnetic field created by a permanent magnet within the speaker. The changing current makes the coil move, which controls a diaphragm to produce sound.